



CTRL80(M2)

Integrated Elevator Controller User Manual



IFE ELEVATORS CO.,LTD



Preface

Thank you for choosing the CTRL80(M2) integrated elevator controller.

The CTRL80(M2) integrated elevator controller is a new-generation integrated elevator controller developed by IFE Elevator Co., Ltd. Based on applications of the CTRL80(M) and combining new industry characteristics, it is the seamless replacement of the CTRL80(M) with complete compatibility in terms of function codes, installation sizes, and control board or lower-level drive. The CTRL80(M2) is easily operable, more energy-saving, safe and comfortable.

This manual describes how to properly use the CTRL80(M2) integrated elevator controller, covering product features, safety information, installation, operation, maintenance and inspection, and troubleshooting. Read this manual thoroughly and keep it properly for future reference.

Contents

Preface	1
Chapter 1 Product Information	4
1.1 Product Overview	4
1.2 Product Features	5
1.3 Designation rules	6
1.4 Mounting Dimensions	6
1.5 Electrical Specifications	7
1.6 Braking Resistor	8
1.7 Terminal Description	9
Chapter 2 Component Description	14
2.1 Operation and Display	14
2.2 CTB Board (IFE-CTB)	20
2.3 HOP Board	22
2.4 CCB Board (IFE-CCB)	29
Chapter 3 System Commissioning	34
3.1 Wiring Diagrams	34
3.2 Description of Shaft Signals	35
3.3 Typical Commissioning	37
Chapter 4 Function Code Table	46
Chapter 5 System Application	74
5.1 Parallel Mode and Group Mode	74
5.2 Emergency Evacuation at Power Failure	75
5.3 Opposite Door Control	79
5.4 Disability Function	82
Chapter 6 Troubleshooting	86
6.1 Description of Fault Levels	86
6.2 Fault Information and Troubleshooting	87



Product Information

Chapter 1 Product Information

1.1 Product Overview

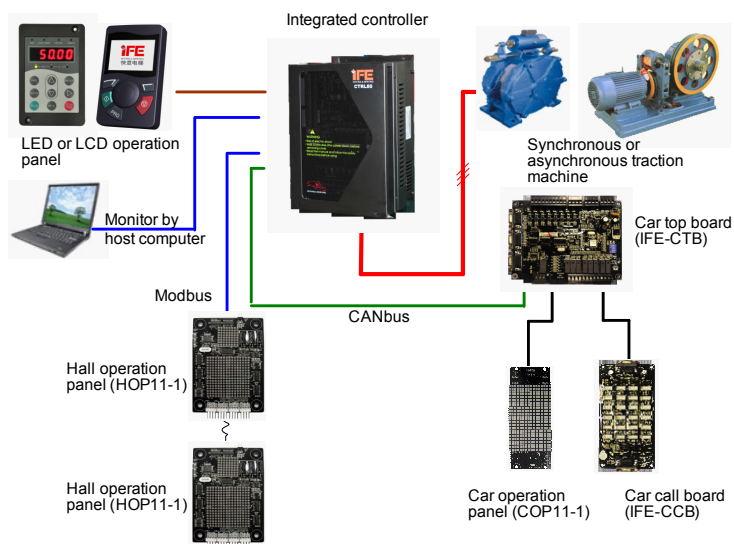
Figure 1-1 Product appearance



The CTRL80 series control system mainly includes the CTRL80 integrated elevator control system, car top board (IFE-CTB), hall operation panel (HOP11-1), car call board (IFE-CCB), and group control board (IFE-GCB).

The following figure shows the system architecture.

Figure 1-2 System architecture of the CTRL80

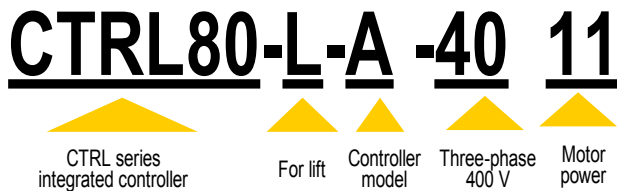


1.2 Product Features

1. More advanced
 - An intelligent control system and being advanced internationally with integrating technologies on computer, automatic control, network communication, and motor vector drive
 - Direct stop technology based on distance control and N curves generated automatically
 - Group control algorithm of less than eight elevators based on fuzzy control theory
 - Multi-CPU redundancy control and supporting CANbus, Modbus, and GSM communication
 - Built-in real-time clock, supporting time-based control to facilitate for intellectualized building management
 - Flexible emergency evacuation operation solutions
 - Automatic identification and operation between short floors
2. Easier to use
 - Compact structure with integration of control and drive, making the equipment room small or unnecessary
 - Easy parameter setting, convenient for commissioning
 - Considerate keypad, making maintenance, inspection and commissioning easier
 - Load cell auto-tuning
 - Multiple commissioning modes: debugger and operation panel
3. Safer and more reliable
 - Multiple safety protections, complying with the GB7588-2003 standards
 - Fault-tolerant design of hardware and software and handling of different types of faults, preventing accidents such as over travel bottom terminal or top terminal
 - Passing professional test by EU MP Laboratory, supporting anti-electromagnetic interference
 - Able to confront power grid fluctuation, dust, high temperature and lightning strike with professional drive manufacture technology and powerful environment adaptation capability
4. More comfortable
 - No-load-cell technology or dedicated load cell compensation device, providing satisfactory startup compensation
 - High-performance vector control, enhancing the riding comfort

1.3 Designation rules

Figure 1-3 Designation rules of the CTRL80(M2)



Mark	A	B	C
Controller Model	CTRL80(M) for asynchronous motor	CTRL80(M) for permanent magnetic synchronous motor (PMSM)	CTRL80(M2) for both PMSM and asynchronous motor

1.4 Mounting Dimensions

The CTRL80(M2) controller has three sizes: SIZE-C, SIZE-D, and SIZE-E, as shown in the following figure. SIZE-F is a customized model with different appearance but the same installation mode.

The following figure and table show the appearance and physical dimensions of the CTRL80(M2) controller.

Figure 1-4 Appearance and mounting dimensions of the CTRL80(M2) controller

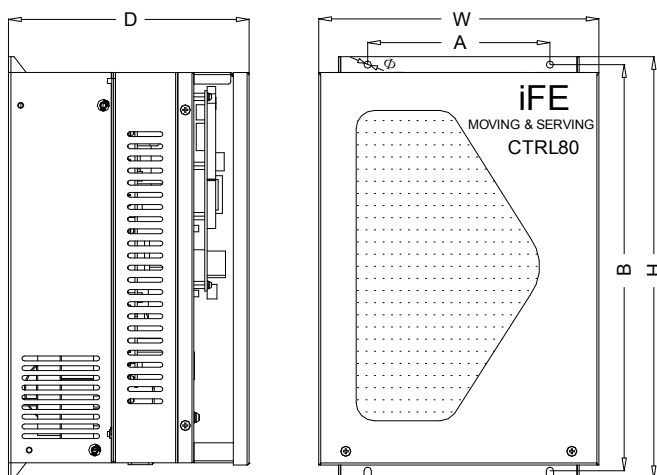


Table 1-1 Sizes of the CTRL80(M2) controller

Size	Model	A (mm)	B (mm)	H (mm)	W (mm)	D (mm)	Hole Diameter (mm)	Gross Weight (kg)
SIZE-C	$P \leq 5.5 \text{ kW}$	140	344	355	224	165	6.5	10
SIZE-D	$5.5 \text{ kW} < P \leq 15 \text{ kW}$	150	335	348	231	198	6.5	12
SIZE-E	$15 \text{ kW} < P \leq 30 \text{ kW}$	235	447	463	285	224	6.5	14.5
SIZE-F	$30 \text{ kW} < P \leq 45 \text{ kW}$	250	598	620	380	262	10	34

Note

The CTRL80(M2) controller of other power ratings, such as above 45 kW is rarely applied in the elevator, and therefore, specifications are not provided here. For further details on the options and availability, please contact IFE.

1.5 Electrical Specifications

Table 1-2 CTRL80(M2) models and electrical specifications

System Model	Power Capacity (kVA)	Input Current (A)	Output Current (A)	Motor Power (kW)
AC supply voltage: single-phase 220 V, range: -15% to 20%				
CTRL80-L-C-2002	2.3	13.2	5.2	1.1
CTRL80-L-C-2003	3.4	17	7.5	1.5
220-CTRL80-L -C-4007	9.8	29	10.3	2.2
220-CTRL80-L -C-4011	12.1	36	15.5	3.7
220-CTRL80-L -C-4015	13.9	41	19	4.0
220-CTRL80-L-C-4018	17.3	40	22.5	5.5
220-CTRL80-L-C-4022	23.1	49	27.7	11
220-CTRL80-L-C-4030	33	61	34.6	15
AC supply voltage: three-phase 220 V, range: -15% to 20%				
CTRL80-L-C-2002	4.0	11.0	9.6	2.2
CTRL80-L-C-2003	5.9	17.0	14.0	3.7
220-CTRL80-L -C-4007	17.0	29.0	18.0	4.0
220-CTRL80-L -C-4011	21.0	36.0	27.0	5.5
220-CTRL80-L -C-4015	24.0	41.0	33.0	7.5
220-CTRL80-L-C-4018	30.0	40.0	39.0	11.0
220-CTRL80-L-C-4022	40.0	49.0	48.0	15.0

System Model	Power Capacity (kVA)	Input Current (A)	Output Current (A)	Motor Power (kW)
220-CTRL80-L-C-4030	57.0	61.0	60.0	18.5
AC supply voltage: three-phase 380 V, range: -15% to 20%				
CTRL80-L-C-4002	4.0	6.5	5.1	2.2
CTRL80-L-C-4003	5.9	10.5	9.0	3.7
CTRL80-L-C-4005	8.9	14.8	13.0	5.5
CTRL80-L-C-4007	11.0	20.5	18.0	7.5
CTRL80-L-C-4011	17.0	29.0	27.0	11.0
CTRL80-L-C-4015	21.0	36.0	33.0	15.0
CTRL80-L-C-4018	24.0	41.0	39.0	18.5
CTRL80-L-C-4022	30.0	49.5	48.0	22.0
CTRL80-L-C-4030	40.0	62.0	60.0	30.0
CTRL80-L-C-4037	57.0	77.0	75.0	37.0
CTRL80-L-C-4045	69.0	93.0	91.0	45.0

Note

Select the proper controller output current based on the rated motor current. Ensure that the controller output current is equal to or greater than the rated motor current.

1.6 Braking Resistor

The models of 30 kW or below have a built-in braking unit, and you only need to connect an external braking resistor between PB and + terminals. For models above 30 kW, you need to install a braking unit and a braking resistor externally.

Select the braking resistor based on the configuration listed in the following table.

Table 1-3 Braking resistor selection for the three-phase 380 V controller

System Model	Average Power of the Braking Resistor (W)	Maximum Resistance (Ω)	Minimum Resistance (Ω)	Braking Unit
CTRL80-L-C-4002	650	230	150	Built-in as standard configuration
CTRL80-L-C-4003	1100	135	100	
CTRL80-L-C-4005	1600	90	80	
CTRL80-L-C-4007	2500	65	50	
CTRL80-L-C-4011	3500	45	35	
CTRL80-L-C-4015	4500	30	25	

System Model	Average Power of the Braking Resistor (W)	Maximum Resistance (Ω)	Minimum Resistance (Ω)	Braking Unit
CTRL80-L-C -4018	5500	25	20	Built-in as standard configuration
CTRL80-L-C -4022	6500	20	15	
CTRL80-L-C -4030	9000	15	15	
CTRL80-L-C -4037	11000	13	12	MDBUN-45-T
CTRL80-L-C -4045	13500	11	11	MDBUN-60-T

Table 1-4 Braking resistor selection for the 220 V controller

System Model	Average Power of the Braking Resistor (W)	Maximum Resistance (Ω)	Minimum Resistance (Ω)	Braking Unit
CTRL80-L-C -2002	650	70	55	Built-in as standard configuration
CTRL80-L-C -2003	1100	40	30	
220-CTRL80-L-C -4007	2500	20	18	
220-CTRL80-L-C -4011	3500	14	10	
220-CTRL80-L-C -4015	4500	10	8	
220-CTRL80-L-C -4018	5500	8	7	
220-CTRL80-L-C -4022	6500	7	6	
220-CTRL80-L-C -4030	9000	7	6	

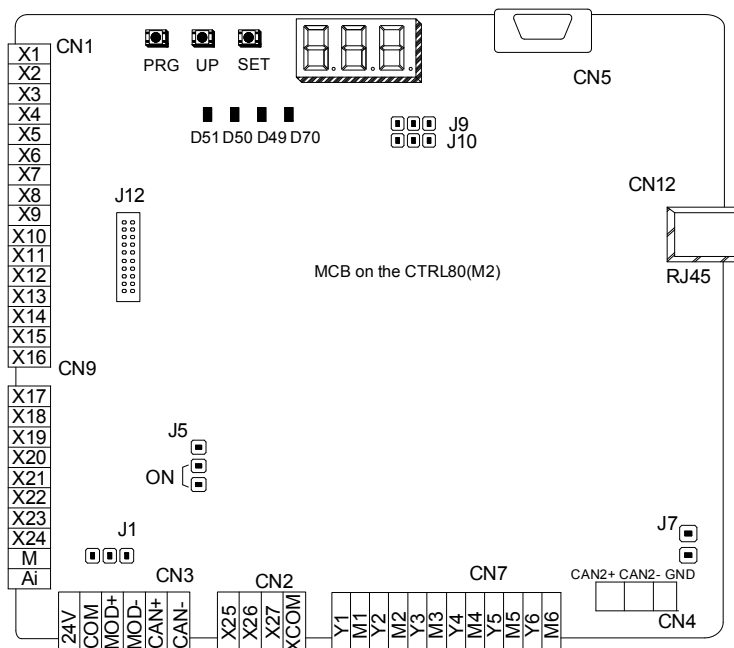
Note

1. The preceding configuration takes the PMSM as an example. The asynchronous motor has poor energy transfer efficiency, and you can reduce the power of the braking resistor or increase the resistance of the braking resistor.
 2. It is recommended that you select the braking resistor closest to the maximum resistance.
-
-

1.7 Terminal Description

Terminals on the main control board (MCB) of the CTRL80(M2) are optimized by adding three strong-current detection terminals, a group of CANbus communication terminals and eliminating the PG card for asynchronous motor. The following figure shows the structure of the MCB.

Figure 1-5 Structure of the MCB on the CTRL80(M2)



The CTRL80(M2) supports two types of external operation panels, universal LED operation panel and LCD operation panel. It provides a keypad with three buttons and three 7-segment LEDs for display, and four LED status indicators. The CTRL80(M2) also supports the external LED operation panel.

Table 1-5 Terminal description and specifications

Mark	Code	Terminal Name	Function Description
CN1	X1 to X16	DI	Input voltage range: 10–30 VDC Input impedance: 4.7 k Ω Optocoupler isolation
CN9	X17 to X24	DI	Input current limit: 5 mA Functions set in F5-01 to F5-24
	Ai/M	AI	Used for the analog load cell device
CN3	24V/COM	External 24 VDC power supply	24 VDC power supply for the entire board
	MOD+/-	RS485 differential signal	Standard isolated RS485 communication interface, used for hall call and display
	CAN+/-	CANbus differential signal	CANbus communication interface, communication with the CTB

Mark	Code	Terminal Name	Function Description
CN2	X25 to X27/ XCOM	Strong-current detection terminal	Input voltage range: 110 VAC $\pm$ 15%, safety and door lock feedback circuit, set in F5-37 to F5-39
CN7	Y1/M1 to Y6/ M6	Relay output	Normally-open (NO), maximum current and voltage rating: 5 A, 250 VAC. Functions set in F5-26 to F5-31
CN4	CAN2+/-	CAN2 bus differential signal	CAN2 communication interface, used for group control or parallel control
CN5	DB9 interface	RS232 communication interface	Used as the interface for commission software, cell/remote monitoring, RS232/RS485 parallel control, and software download for the MCB and drive board
CN12	RJ45 interface	Operation panel interface	Used to connect the digital or LCD operation panel
J1	Optional AI grounding terminal, the pin marked with "COM" being connected to the ground		
J5	Used to connect the terminal resistor for the CANbus communication control board; the pins marked with "ON" being connected to the terminal resistor		
J7	Used to ground the control board. If it is shorted, the ground of the control board is connected to the ground of the drive board.		
J9/ J10	Software writing jumper block (used by the manufacturer). Do not short the pins randomly; otherwise, the controller cannot be used properly.		
J12	Interface for connecting the PG card		

Table 1-6 Description of indicators on the MCB

Mark	Code	Terminal Name	Function Description
D51	COP	CAN1 communication indicator	This indicator is blinking on in green when communication between the MCB and the CTB is normal.
D50	HOP	Modbus communication indicator	This indicator is blinking on in green when communication between the MCB and the HOP is normal.
D49	CAN2	Group control communication indicator	This indicator is steady on in green during parallel control/group control communication, and is blinking on during parallel control/group control running.
D70	232	Serial communication indicator	This indicator is connected to the host computer or cell monitor board, and is on in green when the communication is normal.



2

Component Description

Chapter 2 Component Description

2.1 Operation and Display

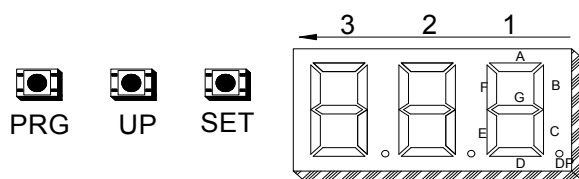
Operations can be performed on the onboard keypad or external operation panel.

2.1.1 Onboard Keypad Description

The onboard keypad consists of three 7-segment LEDs and three buttons. You can query information about the controller and enter simple commands on the keypad.

The following figure shows the appearance of the keypad.

Figure 2-1 Appearance of the keypad

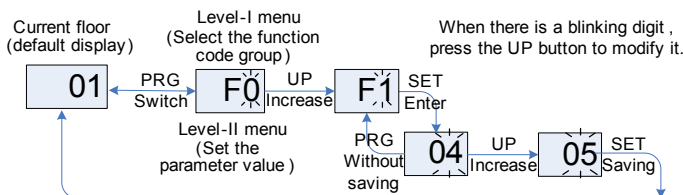


As shown in the preceding figure, the three buttons are PRG, UP, and SET. The functions of the three buttons are as follows:

- **PRG:** Press this button in any state to display the current function group number.
You can press the UP button to change the function group number.
- **UP:** Press this button to increase the function group number.
Currently, the MCB defines a total of 13 function code groups, namely, F0 to F9, FA to FC. You can press the UP button to display them in turn. In addition, in special function code group menu, you can input simple references by using the UP button.
- **SET:** In the function code group menu, press this button to enter the menu of the function code group.
In special function code group menu, after you input a simple reference and press this button to save the setting, the keypad enters the F0 menu by default.

The following figure shows the setting of increasing the called floor to 5.

Figure 2-2 Setting the called floor



The function code groups displayed on the keypad are described as follows:

1. F0: display of floor and running direction

The F0 menu is displayed on the keypad by default upon power-on. The first 7-segment LED indicates the running direction, while the last two 7-segment LEDs indicate the current floor of the elevator.

When the elevator stops, the first 7-segment LED has no display. When the elevator runs, the first 7-segment LED indicates the running direction.

When a system fault occurs, the 7-segment LEDs automatically display the fault code and blink. If the fault is reset, the F0 menu is displayed.

2. F1: command input of the running floor

After you enter the F1 menu, the 7-segment LEDs display the bottom floor (F6-01). You can press the UP key to set the destination floor within the range of lowest to top and then press SET to save the setting. The elevator runs to the destination floor, and the 7-segment LEDs automatically switch over to the F0 menu at the same time.

3. F2: fault reset

After you enter the F2 menu, the 7-segment LEDs display "0". When you press UP to select 1 and press SET, the system fault is reset. After the system fault is reset, the 7-segment LEDs automatically switch over to the F0 menu.

The value 2 indicates to display the fault code and time. After you select this value and press SET, the system displays the fault codes and time of 11 fault records in turn. Press PRG to exit.

4. F3: time display

After you enter the F3 menu, the 7-segment LEDs display the current system time circularly.

5. F4: contract number display

After you enter the F4 menu, the 7-segment LEDs display the user's contract number.

6. F5: running times display

After you enter the F5 menu, the 7-segment LEDs display the elevator running times circularly.

7. F6: door open/close control

After you enter the F6 menu, the 7-segment LEDs display "1-1" by default. In this case, the UP and SET buttons can be respectively used for door open and door close. Press PRG to exit.

8. F7: shaft auto-tuning command input

After you enter the F7 menu, the 7-segment LEDs display "0". You can select 0 or 1 here, where 1 indicates the shaft auto-tuning command available. After you select 1 and press SET, shaft auto-tuning is implemented if the conditions are met. Meanwhile, the 7-segment LEDs switch over to the F0 menu. After shaft auto-tuning is complete, F7 is back to "0" automatically. If shaft auto-tuning conditions are not met, fault code "E35" is displayed.

9. F8: test function

After you enter the F8 menu, the 7-segment LEDs display "0". The setting range of F8 is 1–4, described as follows:

- 1: Hall call forbidden
- 2: Door open forbidden
- 3: Overload forbidden
- 4: Limit switches disabled

After the setting is complete, press SET. Then the 7-segment LEDs display "E88" and blink, prompting that the elevator is being tested. When you press PRG to exit, F8 is back to "0" automatically.

10. F9: Reserved

11. FA: auto-tuning function

After you enter the FA menu, the 7-segment LEDs display "0". The setting range of FA is 0–2, described as follows:

- 1: With-load auto-tuning
- 2: No-load auto-tuning

After the setting is complete, press SET. Then, the 7-segment LEDs display "TUNE", indicating that the elevator enters the auto-tuning state. After ensuring that the safety running conditions are met, press SET again to start auto-tuning. After the auto-tuning is complete, the 7-segment LEDs display the current angle for 2s, and then switch over to the F0 menu. Press PRG to exit the auto-tuning state.

12. FB: CTB status display

After you enter the FB menu, the 7-segment LEDs display the I/O status of the CTB. The meaning of each segment is listed in the following table.

Table 2-1 LED display of the CTB status

NO.	Mark	Function	Meaning of ON
1	A	Light curtain 1	Light curtain 1 input active
	B	Light curtain 2	Light curtain 2 input active
	C	Door open limit 1	Door open limit 1 input active
	D	Door open limit 2	Door open limit 2 input active
	E	Door close limit 1	Door close limit 1 input active
	F	Door close limit 2	Door close limit 2 input active
	G	Full-load signal	Full-load signal active
	DP	Over-load signal	Over-load signal active
2	A	Light-load signal	Light-load signal active
3	A	Door open output 1	Door open 1 relay output
	B	Door close output 1	Door close 1 relay output
	C	Forced door close 1	Forced door close 1 relay output
	D	Door open output 2	Door open 2 relay output
	E	Door close output 2	Door close 2 relay output
	F	Forced door close 2	Relay output of forced door close 2
	G	Up arrival gong	Relay output of up arrival gong
	DP	Down arrival gong	Relay output of down arrival gong

13. FC: elevator direction change

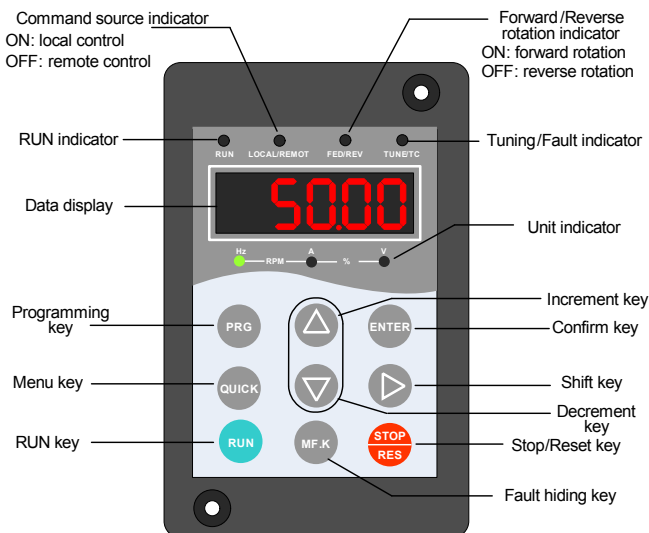
This menu has the same function as F2-10. The values are as follows:

- 0: Running direction and position pulse direction unchanged
- 1: Running direction reversed, position pulse direction reversed
- 2: Running direction unchanged, position pulse direction reversed
- 3: Running direction reversed, position pulse direction unchanged

2.1.2 Use of the LED Operation Panel

The LED operation panel is connected to the RJ45 interface of the CTRL80 controller by means of an 8-core flat cable. You can modify the function parameters, monitor the working status, and run or stop the CTRL80 controller by using the LED operation panel. The following figure shows the appearance of the LED operation panel.

Figure 2-3 Appearance of the LED operation panel



■ Function Indicators

- **RUN**: This indicator is on when the CTRL80(M2) is running.
- **LOCAL/REMOT**: Reserved.
- **FWD/REV**: It is the elevator direction indicator. ON indicates down direction, and OFF indicates up direction.
- **TUNE/TC**: It is the auto-tuning indicator. This indicator is on when the elevator is in auto-tuning state.

■ Unit Indicators

○ indicates OFF, and ● indicates ON.

Hz ●—RPM—○—%—○ Hz: unit of frequency

Hz ○—RPM—●—%—○ A: unit of current

Hz ○—RPM—○—%—● V: unit of voltage

Hz ●—RPM—●—%—○ RPM: unit of rotational speed

Hz ○—RPM—●—%—● %: percentage

■ Keys on the Operation Panel

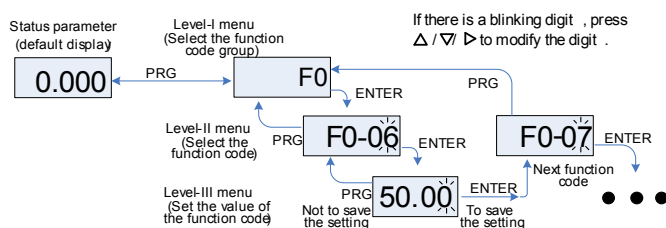
Table 2-2 Keys on the operation panel

Key	Name	Function
	Programming	Enter or exit Level I menu.
	Confirm	Enter the menu interfaces level by level, and confirm the parameter setting.
	Increment	Increase data or function code.
	Decrement	Decrease data or function code.
	Shift	Select the displayed parameters in turn in the stop or running state, and select the digit to be modified when modifying parameters.
	RUN	In operation panel control mode, press this key to run the controller.
	Stop/Reset	In operation panel control mode, press this key to stop the running or reset the controller in fault state.
	Quick	Enter or exit Level I shortcut menu.
	Fault hiding	Press this key to display or hide fault information in fault state.

■ Operations on the Three-Level Menu

The following figure shows the operation procedure on the three-level menu.

Figure 2-4 Operation procedure on the three-level menu



2.2 CTB Board (IFE-CTB)

The car top board (IFE-CTB) is the elevator car control board of the CTRL80 series controller. It consists of eight DI terminals, one AI terminal, and nine relay output terminals. The CTB can communicate with the CCB and HOP. The following figure shows the structure and size of the CTB.

Figure 2-5 Structure and size of the CTB

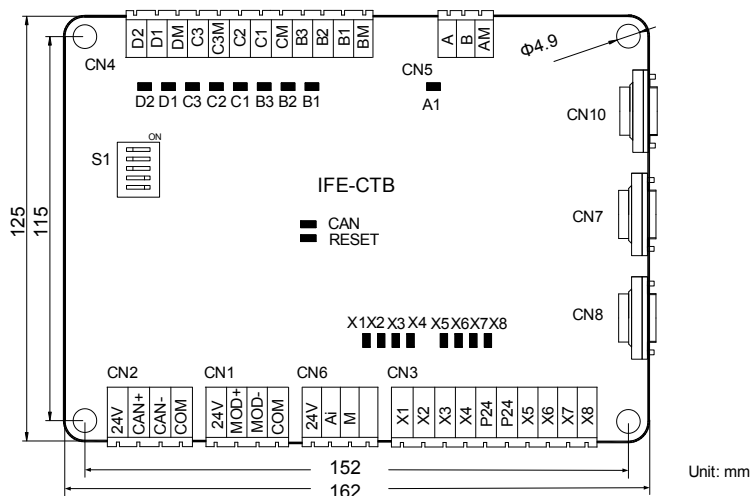


Table 2-3 Description of CTB terminals

Type	Mark	Terminal Name	Description
CN2 (communication with the MCB)	+24V/COM	External 24 VDC power supply	24 VDC power supply to the entire board
	CAN+/CAN-	CAN communication	Connecting the MCB
CN1 (communication with the HOP)	+24V/COM	24 VDC power supply	24 VDC power supply to the HOP
	MOD+/MOD-	Modbus communication	Connecting the HOP
CN6 (AI)	Ai-M	Load cell signal input	0–10 VDC

Type	Mark	Terminal Name	Description
CN3 (DI)	P24	24 VDC power supply	DI common terminal
	X1	Light curtain 1	1. Photocoupler isolation, unipolarity input 2. Input impedance: 3.3 k Ω 3. IFE-CTB signal valid with 24 VDC power supply
	X2	Light curtain 2	
	X3	Door open limit 1	
	X4	Door open limit 2	
	X5	Door close limit 1	
	X6	Door close limit 2	
	X7	Full-load signal	
	X8	Overload signal	
CN4 (relay output)	B1-BM	Door open signal 1	30 VDC, 1 A
	B2-BM	Door close signal 1	
	B3-BM	Forced door close 1	
	C1-CM	Door open signal 2	
	C2-CM	Door close signal 2	
	C3-C3M	Forced door close 2	
	D1-DM	Up arrival signal	
	D2-DM	Down arrival signal	
CN5	A-AM	Car fan and lamp control	250 VAC, 3 A; 30 VDC, 1 A
CN7/8	DB9 pin port	CCB communication	Connecting the CCB
CN10	DB9 pin port	Host computer communication	Connecting the host computer, burning software
CAN	Indicator	Communication indicator	This indicator blinks when the CTB communicates with the MCB.
RESET	Indicator	Fault indicator	This indicator blinks and the CAN indicator is steady on when a fault occurs during communication between the CTB and the MCB.

Table 2-4 Description of DIP switch

Name	DIP Bit	Function
S1	1	Used in parallel control, set to ON when connecting to the CTB of elevator 2.
	2	Reserved
	3	Reserved
	4	Setting the terminal resistor for Modbus communication
	5	Setting the terminal resistor for CANbus communication

Note

To prevent external interference on the communication, use the shielded twisted pairs as the communication cable, do not lay the cables in parallel, and connect the cables to the correct terminals securely according to the terminal mark.

2.3 HOP Board

As an important interface between users and the control system, the HOP receives hall calls and displays the current floor and running direction for the hall. This board can also be used as car operation panel.

IFE provides many types of display boards. The following table describes only a few common types. If you need other types, please contact IFE.

The common types to be described are listed in the following table.

Table 2-5 Common types of HOP

Name	Feature	Size (mm)
HOP11-1-IF2	Dot-matrix display board (orange)	100 x 70 x 10
COP11-1-G2	Large dot-matrix car display board (orange)	157 x 65 x 22
HOP11-1-B	No-display serial-parallel conversion board	84 x 70 x 20

2.3.1 HOP11-1-IF2 (Hall Dot-Matrix Display Board)

The following figures show the appearance and size of HOP11-1-IF2.

Figure 2-6 Appearance of HOP11-1-IF2

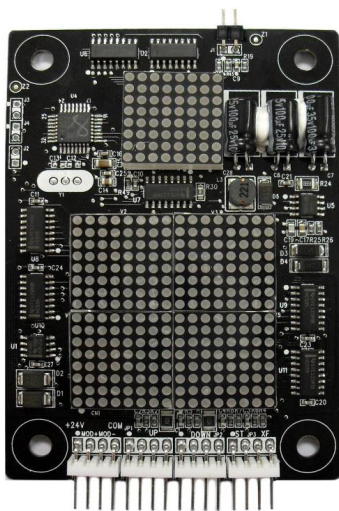


Figure 2-7 Size of HOP11-1-IF2

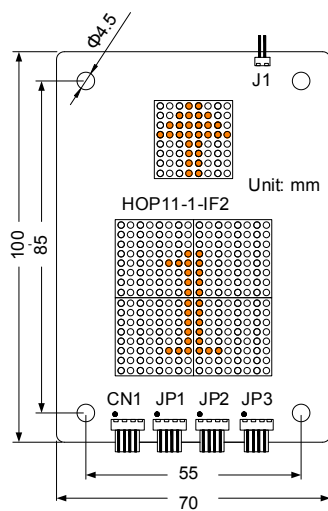
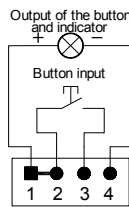


Table 2-6 Input and output terminals of HOP11-1-IF2

Terminal Name	Function	Description
JP1	Interface for the up call button and indicator	Pins 2 and 3 are for call input and become ON when the button is pressed. Pins 1 and 4 are for control of the down call indicator. 
JP2	Interface for the down call button and indicator	
Note: Pins 1 and 2 are connected to the positive of 24 VDC power supply. The pin with white dot or that is rectangular is pin 1.		
JP3	Interface for the fire emergency switch and elevator lock switch	Pins 1 and 2 are for elevator lock signal input; when pins 1 and 2 become ON, the elevator lock signal is output. Pin 3 and 4 are for fire emergency signal input; when pins 3 and 4 become ON, the fire emergency signal is output.
J1	Interface for setting the floor address	Short J1, and press the UP button or DOWN button to set the floor address. After the jumper cap is removed, the address is automatically stored.
CN1	Interface for Modbus communication and power supply	Pins 1 and 4 are respectively connected to 24 VDC and 0 V power supply. Pins 3 and 4 are respectively connected to MOD+ and MOD- terminals.

2.3.2 COP11-1-G2 (Large Dot-Matrix Car Display Board)

The following figures show the appearance and size of COP11-1-G2.

Figure 2-8 Appearance of COP11-1-G2

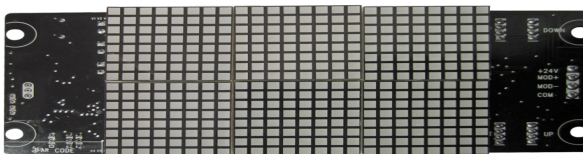


Figure 2-9 Size of COP11-1-G2

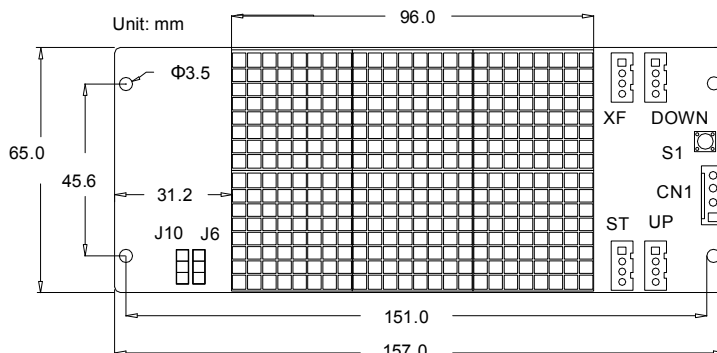


Table 2-7 Input and output terminals of COP11-1-G2

Terminal Name	Function	Description
ST	Interface for the elevator lock switch	Pins 2 and 3 are for elevator lock signal input; when pins 2 and 3 become ON, the elevator lock signal is output. Pins 1 and 4 are for the elevator lock indicator control.
XF	Interface for the fire emergency switch	Pins 2 and 3 are for fire emergency signal input; when pins 2 and 3 become ON, the fire emergency signal is output. Pins 1 and 4 are for the fire emergency indicator control.
UP	Interface for the up call button and indicator	Pins 2 and 3 are for button signal input; pins 2 and 3 become ON when the button is pressed. Pins 1 and 4 are for control of the up/down call indicator.
DOWN	Interface for the down call button and indicator	
Note: Pins 1 and 2 are connected to the positive of 24 VDC power supply. The pin with white dot or that is rectangular is pin 1.		
S1	Interface for setting the floor address	Hold down S1 to set the floor address. After you release S1, the address blinks three times and is stored.
CN1	Interface for Modbus communication and power supply	Pins 1 and 4 are respectively connected to 24 VDC and 0 V power supply. Pins 3 and 4 are respectively connected to MOD+ and MOD- terminals.
J6	Interface for selecting English/Chinese	The display is in English if pins 1 and 2 are shorted, and in Chinese if pins 2 and 3 are shorted.
J10	Interface for selecting horizontal/vertical display	The display is vertical if pins 1 and 2 are shorted, and horizontal if pins 2 and 3 are shorted. The display is horizontal by default.

2.3.3 HOP11-1-B (No-display Serial-Parallel Conversion Board)

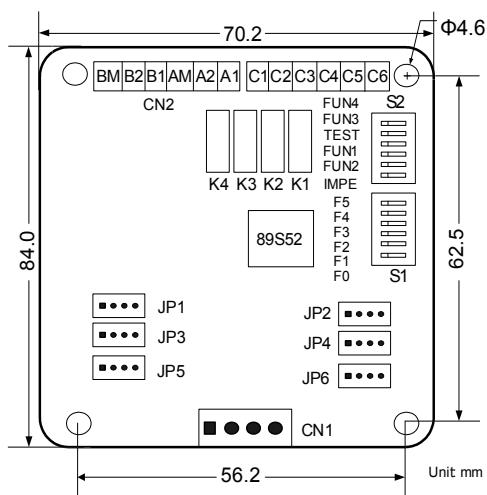
■ Appearance and Size

The following figures show the appearance and size of HOP11-1-B.

Figure 2-10 Appearance of HOP11-1-B



Figure 2-11 Size of HOP11-1-B



■ Terminal Description

Table 2-8 DIP switch description

S1	1 to 5	Floor Address Setting (Range: 0–63)
S1.1	F0	Set the floor address in binary mode.
S1.2	F1	
S1.3	F2	
S1.4	F3	
S1.5	F4	
S1.6	F5	
S2	1 to 5	Function Selection
S2.1	IMPE	Setting the terminal resistor for Modbus communication
S2.2	FUN2	Function selection
S2.3	FUN1	Function selection
S2.4	TEST	For the test
S2.5	FUN3	Function selection
S2.6	FUN4	Function selection

Table 2-9 Input and output terminals of HOP11-1-B

Terminal Name	Function	Pin Description			
		1	2	3	4
JP1	Interface for elevator lock switch input	+24V	+24V	Elevator locking signal input	Elevator locking indicator output
JP2	Interface for the fire emergency switch	+24V	+24V	Fire emergency signal input	Fire emergency indicator output
JP3	Interface for the up call button and indicator	+24V	+24V	Up call button input	Up call indicator output
JP4	Interface for the down call button and indicator	+24V	+24V	Down call button input	Down call indicator output
JP5	Interface for the up call button and indicator for the disabled	+24V	+24V	Up call button input	Up call indicator output
JP6	Interface for the down call button and indicator for the disabled	+24V	+24V	Down call button input	Down call indicator output

Terminal Name	Function	Pin Description			
		1	2	3	4
CN1	Interface for Modbus communication and power supply	+24V	MOD+	MOD-	COM
CN2	Relay output terminal	See the following part "Function Description".			
CN3	Open-collector output terminal	See the following part "Function Description".			
The pin with white dot or that is rectangular is pin 1. The other pins are pins 2, 3, and 4 in order.					

■ Function Description

The function of HOP11-1-B can be selected by means of four DIP switches (FUN1, FUN2, FUN3, and FUN4) on S2. The functions are described in the following table.

Table 2-10 Functions available for HOP11-1-B

	S2.6, FUN4	S2.5, FUN3	S2.3, FUN1	S2.2, FUN2
No-display HOP	OFF	OFF	ON	OFF
7-segment function	OFF	OFF	OFF	OFF
Binary output	-	-	-	ON
Note: When the binary output function is used, you only need to set FUN2 to ON.				

1. No-display HOP

The no-display HOP has four relay outputs, K1, K2, K3, and K4, provided by CN2. The following table lists the input and output terminals and function. For details, see the corresponding terminal description.

Relay Name	Corresponding CN2 Pin	Common Pin	Function
K1	A1	AM	Up arrival indicator
K2	A2	AM	Down arrival indicator
K3	B1	BM	Up arrival gong
K4	B2	BM	Down arrival gong

2. 7-segment Function

Corresponding CN2 Pin	Common Pin	Function
A1	AM	Overload/Full-load
A2	AM	Segment G
B1	BM	Up indication
B2	BM	Down indication
Corresponding CN3 Pin	Common Pin	Function
C1	BM	Segment A
C2	BM	Segment B
C3	BM	Segment C
C4	BM	Segment D
C5	BM	Segment E
C6	BM	Segment F

3. Binary Output

When this function is used, C1 to C6 respectively output the binary bits, displaying combinations of all numbers and letters. For details, see the descriptions of group FE.

2.4 CCB Board (IFE-CCB)

The car call board (IFE-CCB) is another interface between users and the control system. Each CCB comprises 24 inputs and 22 outputs, including 16 floor buttons and 8 functional signals. The CCB mainly collects button calls and outputs signals of the button call indicators. The need for 31-floor use can be implemented through cascaded connection. CN2 is an input connector and CN1 is a cascaded output connector.

The following figures show the appearance and structure of the CCB.

Figure 2-12 Appearance of the CCB

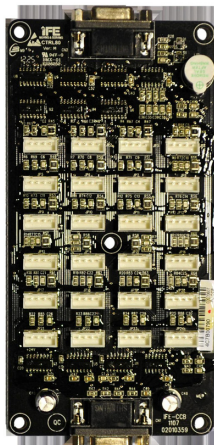


Figure 2-13 Structure of the CCB

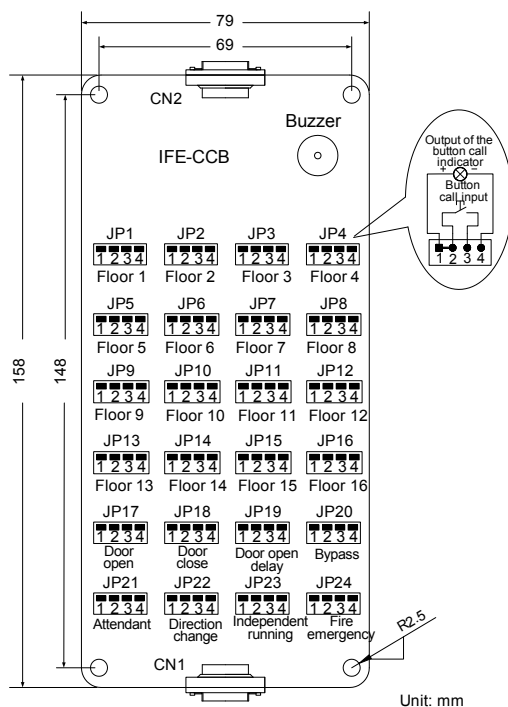


Table 2-11 Input and output terminals of the CCB

No.	Interface	Pins 2 and 3	Pins 1 and 4	Remarks
1	JP1	Floor 1 button input	Floor 1 display output	For CCB2, the input signal of JPn corresponds to floor (16+n) button input (1 ≤ n ≤ 16).
2	JP2	Floor 2 button input	Floor 2 display output	
3	JP3	Floor 3 button input	Floor 3 display output	
4	JP4	Floor 4 button input	Floor 4 display output	
5	JP5	Floor 5 button input	Floor 5 display output	
6	JP6	Floor 6 button input	Floor 6 display output	
7	JP7	Floor 7 button input	Floor 7 display output	
8	JP8	Floor 8 button input	Floor 8 display output	
9	JP9	Floor 9 button input	Floor 9 display output	
10	JP10	Floor 10 button input	Floor 10 display output	
11	JP11	Floor 11 button input	Floor 11 display output	
12	JP12	Floor 12 button input	Floor 12 display output	
13	JP13	Floor 13 button input	Floor 13 display output	
14	JP14	Floor 14 button input	Floor 14 display output	
15	JP15	Floor 15 button input	Floor 15 display output	
16	JP16	Floor 16 button input	Floor 16 display output	
17	JP17	Door open button input	Door open display output	Invalid for CCB 2.
18	JP18	Door close button input	Door close display output	
19	JP19	Door open delay button input	Door open delay display output	
20	JP20	Bypass input	Bypass display output	
21	JP21	Attendant input	Reserved	
22	JP22	Direction change input	Reserved	
23	JP23	Independent running input	Reserved	
24	JP24	Firefighter input	Reserved	
Note: Pins 1 and 2 are connected to the positive of power supply. The pin with white dot mark or that is rectangular is pin 1.				



3

System Commissioning

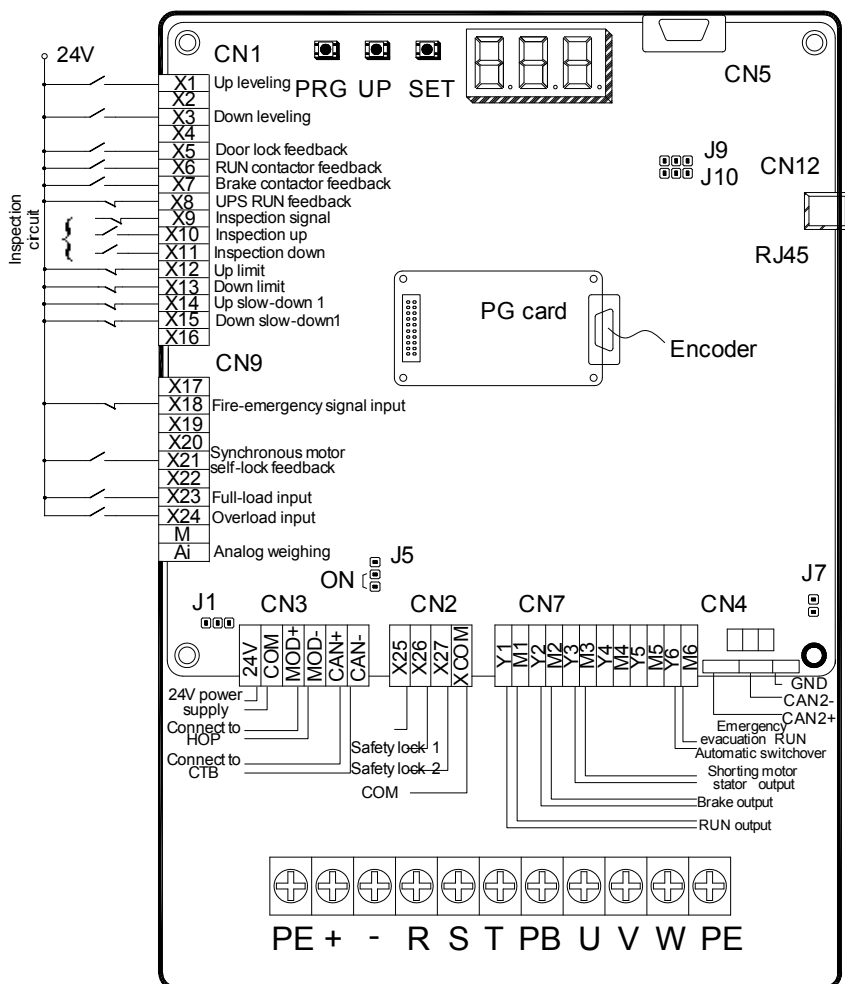
Chapter 3 System Commissioning

3.1 Wring Diagrams

3.1.1 Electric Wiring

The following figure shows the electric wiring diagram of the CTRL80(M2).

Figure 3-1 Electric wiring diagram of the CTRL80(M2)



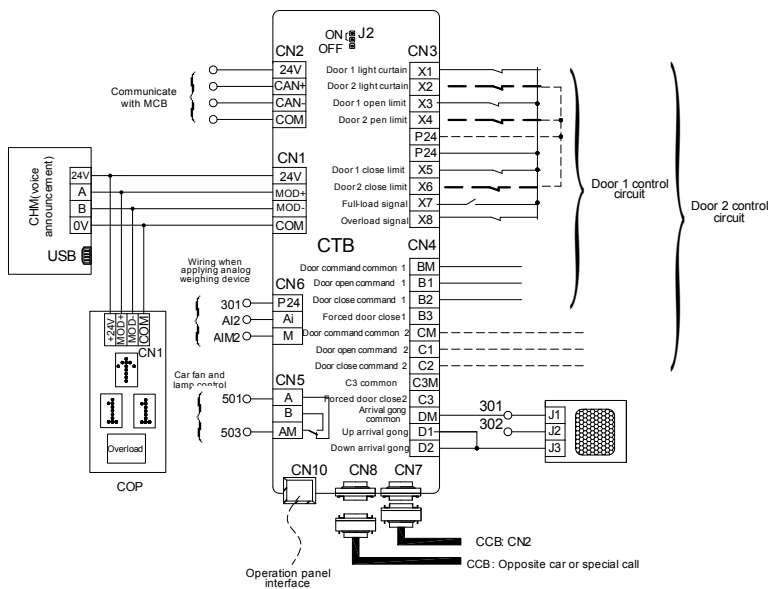
Note

The above wiring diagram shows the default I/O function setting of group F5. The braking resistor is connected to PB and + terminals.

3.1.2 Recommended CTB Wiring and Parameter Setting

The following wiring diagram shows the default function setting of the CTB. You can change the input polarity (NO/NC) of the CTB terminals in F5-25.

Figure 3-2 Recommended CTB wiring



3.2 Description of Shaft Signals

In elevator control, the car position needs to be identified based on shaft position signals, implementing stop at floors accurately and guaranteeing safe running.

These shaft position signals include the up final limit switch, down final limit switch, up limit switch, down limit switch, up slow-down switch, down slow-down switch and leveling switches.

3.2.1 Limit Switch and Final Limit Switch

Stop the car at the top (or bottom) leveling position, and mount the limit switch 100 mm away from the top (or bottom) leveling position. That is, the limit switch shall act when the car continues to run upward (or downward) 100 mm from the top (or bottom) leveling position.

The final limit switch is mounted above the up limit switch or below the down limit switch. It is usually 100 to 150 mm away from the top (or bottom) leveling position.

3.2.2 Slow-Down Switch

Slow-down switch is one of the key protective components of the CTRL80 series controller, protecting the elevator from over travel top terminal or over travel bottom terminal at maximum speed when the elevator position becomes abnormal.

The default deceleration rate ("Special deceleration rate" in F3-08) of the CTRL80 series controller is 0.9 m/s^2 . In the condition that the rated speed and the special deceleration rate are certain, the slow-down distance can be obtained via the following formula:

Slow-down distance = Rated speed $\times$ Rated speed / ($2 \times$ Special deceleration rate)

Then you can obtain the mounting positions of the slow-down switches that correspond to various elevator speeds.

The recommended positions of slow-down switches are listed in the following table.

Table 3-1 Recommended positions of slow-down switches

Rated Elevator Speed	$V \leq 1.75 \text{ m/s}$	$1.75 \text{ m/s} < V \leq 2.5 \text{ m/s}$	$2.5 \text{ m/s} < V \leq 3.7 \text{ m/s}$
Slow-down switch 1	1.3 m to L/2	1.3 m	1.3 m
Slow-down switch 2		3.7 m	3.7m
Slow-down switch 3			8.0 m

Note

- "L" in the table indicates the landing height. The default special deceleration rate is 0.9 m/s^2 , and the acceleration rate and deceleration rate are 0.6 m/s^2 .
- The positions of slow-down switches are calculated on the basis of the default value (special deceleration rate 0.9 m/s^2 , and acceleration rate and deceleration rate 0.6 m/s^2). Decreasing the acceleration rate and deceleration rate or increasing the special deceleration rate does not affect the safety, but increasing the acceleration rate and deceleration rate or decreasing the special deceleration rate may bring safety hazard. If any value needs to be changed, calculate proper slow-down distance for mounting the slow-down switches.

3.2.3 Leveling Switch

The CTRL80 system allows the use of one to three leveling switches. If only one leveling sensor is used, the corresponding function code which is used for input of leveling signal is allocated with function 03 (NO input of door zone signal).

There is no special requirement on the length of the flag. It is recommended to use a flag of 200 mm long. Ensure that flags at all landings have the same length and are mounted vertically.

You need to add two re-leveling switches if applying the door pre-open function. In this case, you need to increase the length of the flag properly. For details on the door pre-open module, contact IFE for more information.

3.3 Typical Commissioning

3.3.1 Commissioning at Inspection Speed

The CTRL80(M2) controller supports both asynchronous motor and PMSM (switched over by means of F1-25).

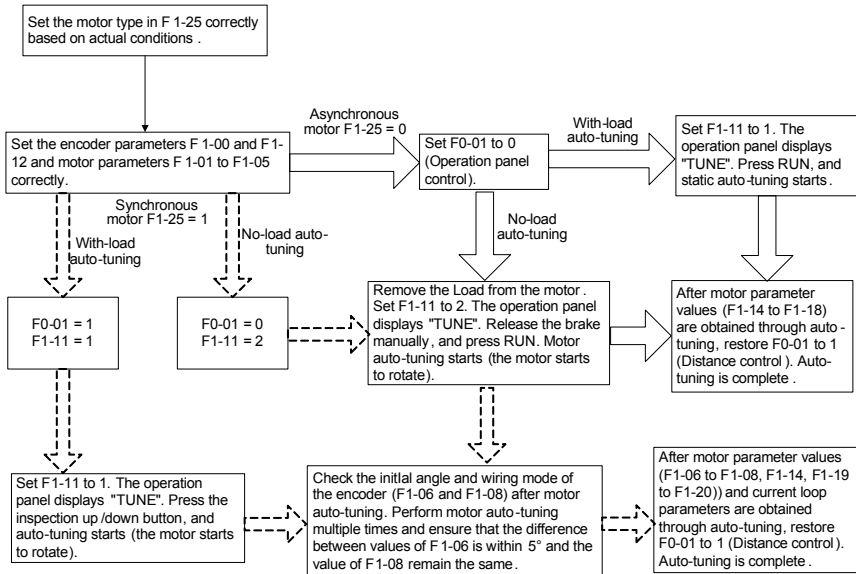
To implement accurate control on the motor, motor auto-tuning must be performed before use of the CTRL80(M2).

Follow the following instructions on the motor auto-tuning:

1. Ensure that the installation and wiring satisfy the safety requirements and technical specifications before the auto-tuning.
2. Set the motor type (F1-25) and enter the motor parameters in group F1 (F1-01 to F1-05) correctly. Otherwise, auto-tuning may not be performed.
3. Set the encoder type (F1-00) and pulses per revolution of the encoder (F1-12) correctly. Set F0-00 to 1 (CLVC) and F0-01 to 1 (Distance control) before auto-tuning.
4. Set F8-01 to 0 (disabling the pre-torque). Otherwise, an abnormality may occur during auto-tuning.
5. For with-load auto-tuning, ensure that the motor wiring is correct (UVW wires of the motor respectively connected to UVW wires of the controller). If the motor wiring is incorrect, the motor may jitter or fail to run after the brake is released. If such error occurs, replace any two of UVW wires.
6. Reset the fault if the system is in fault state before auto-tuning, because the system cannot enter the auto-tuning state (does not display "TUNE") in such state.
7. Perform auto-tuning again on the PMSM if the wiring or encoder of the motor is changed.
8. Perform auto-tuning multiple times (more than three) for the PMSM, compare the PG card initial angles (F1-06). If the error is within 5°, auto-tuning is successful.
9. Perform trial running after auto-tuning is completed, and check whether the current is normal and whether the running direction is the same as the set direction. If not, change value of F2-10.
10. Ensure that no person is in the shaft during with-load auto-tuning because it is dangerous.

The following figure shows the motor auto-tuning process.

Figure 3-3 Motor auto-tuning process



For the PMSM, in with-load auto-tuning, stator resistance, shaft-D and shaft-Q inductance, current loop (including zero servo) PI parameters, and encoder initial angle are learned. In no-load auto-tuning, the encoder wiring mode is also learned.

For the asynchronous motor, in static auto-tuning, stator resistance, rotor resistance, and leakage inductance are learned, and the mutual inductance and motor magnetizing current are calculated automatically. In complete auto-tuning, the mutual inductance, motor magnetizing current, and current loop parameters are learned.

3.3.2 Commissioning at Normal Speed

After ensuring that running at inspection speed is normal, you can start commissioning at normal speed. Shaft auto-tuning is required before the commissioning at normal speed. Ensure that the elevator satisfies the safety running requirements.

To perform shaft auto-tuning, the following conditions must be satisfied:

1. The signals of the encoder and leveling sensors (NC, NO) are correct and the position switches are installed properly.
2. The elevator is at the bottom floor and the down slow-down switch 1 acts.
3. The elevator is in the inspection state. The control mode is distance control and CLVC (F0-00 = 1, F0-01 = 1).
4. The top floor number (F6-00) and bottom floor number (F6-01) are set correctly.
5. The CTRL80 system is not in the fault alarm state. If there is a fault at the moment, press  to reset the fault.

Then set F1-11 to 3 on the operation panel or set F7 to 1 on the keypad of the MCB, and start shaft auto-tuning.

3.3.3 Riding Comfort

The riding comfort is an important factor of the elevator's overall performance. Improper installation of mechanical parts and improper parameter settings will cause bad comfort. Enhancing the riding comfort mainly involves adjustment of the controller output and the elevator's mechanical construction.

■ Controller Output

The parameters that may influence the riding comfort are described in this part.

Function Code	Parameter Name	Setting Range	Default	Description
F1-09	Current filter time	0.00–40.00	0.00	It can reduce the lower-frequency vertical jitter during running.
F1-14	Stator resistance	00.000–30.000 Ω	0.00 Ω	Decreasing the values of these parameters slightly for the asynchronous motor can improve the riding comfort.
F1-15	Rotor resistance	00.000–30.000 Ω	0.00 Ω	
F1-18	Magnetizing current	0.01–300.00 A	0.00 A	Increasing the value of this parameter can improve the load capacity of the asynchronous motor.
F2-00	Speed loop proportional gain 1	0–100	40	F2-00 and F2-01 are the PI regulation parameters when the running frequency is lower than F2-02 (Switchover frequency 1). F2-03 and F2-04 are the PI regulation parameters when the running frequency is higher than F2-02 (Switchover frequency 2). The regulation parameters between F2-02 and F2-04 are the weighted average value of F2-00 & F2-01 and F2-03 & F2-04.
F2-01	Speed loop integral time 1	0.01–10.00s	0.60s	
F2-02	Switchover frequency 1	0.00 to F2-05	2.00 Hz	
F2-03	Speed loop proportional gain 2	0–100	35	
F2-04	Speed loop integral time 2	0.01–10.00s	0.80s	
F2-05	Switchover frequency 2	F2-02 to F0-06	5.00 Hz	

For a faster system response, increase the proportional gain and reduce the integral time. Be aware that a fast system response causes system oscillation.

The recommended regulating method is as follows:

If the factory parameters cannot satisfy the requirements, make slight regulation. Increase the proportion gain first to the largest value under which the system does not oscillate. Then decrease the integral time to ensure fast responsiveness and small overshoot.

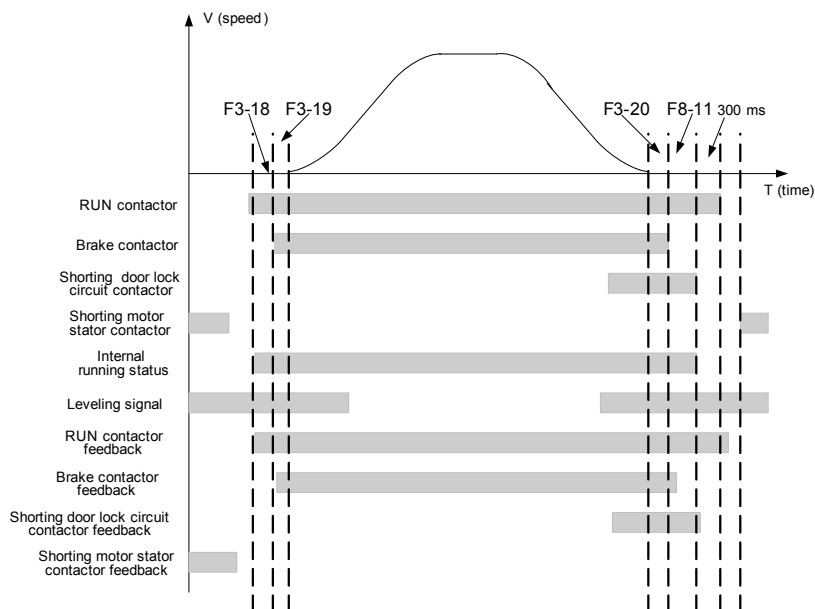
If both F2-02 (Switchover frequency 1) and F2-05 (Switchover frequency 2) are set to 0, only F2-03 and F2-04 are valid.

Function Code	Parameter Name	Setting Range	Default	Description
F2-06	Current loop proportion gain	10–500	60	F2-06 and F2-07 are the current loop adjustment parameters in the vector control algorithm.
F2-07	Current loop integral gain	10–500	30	

Regulating the two parameters for PMSM has obvious effect on the riding comfort. Appropriate regulation can restrain jitter during running. Generally, increase their default values to 180 and 60 respectively for the PMSM.

Function Code	Parameter Name	Setting Range	Default	Description
F3-00	Startup speed	0.000–0.030 m/s	0.010 m/s	It can reduce the terrace feeling at startup caused by the breakout friction of guide rail.
F3-01	Holding time	0.000–0.500s	0.150s	
F3-18	Zero-speed control time at startup	0.000–1.000s	0.200s	It specifies the zero speed holding time before brake output.
F3-19	Brake release delay	0.000–1.000s	0.200s 0.600s	It specifies the time required from when the system outputs the open signal to when the brake is completely released. The value is automatically changed to 0.0600s by the system when F8-01 is 2 (Automatic pre-torque compensation).
F3-20	Zero-speed control time at end	0.200–1.500s	0.200s	It specifies the zero speed holding time after the brake is applied.
F8-11	Brake apply delay	0.000–1.000s	0.300s	It specifies the time from when the system outputs the close signal to when the brake is completely applied.

Figure 3-4 Running time sequence



The release time of the brakes varies according to the types and the response time of the brakes is greatly influenced by the ambient temperature. Too high brake coil temperature slows the brake responsiveness. Thus, when the riding comfort at startup or stop cannot be improved by adjusting zero servo or load cell compensation parameters, appropriately increase the values of F3-19 and F3-20 to check whether the brake release time influences the riding comfort.

Function Code	Parameter Name	Setting Range	Default	Description
F8-01	Pre-torque selection	0: Pre-torque invalid 1: Load cell pre-torque compensation 2: Automatic pre-torque compensation	0	None.
F2-11	Zero servo current coefficient	0.20%–50.0%	15.0%	These are zero servo adjustment parameters (automatic pre-torque compensation).
F2-12	Zero servo speed loop KP	0.00–2.00	0.50	
F2-13	Zero servo speed loop TI	0.00–2.00	0.60	

Function Code	Parameter Name	Setting Range	Default	Description
F8-02	Pre-torque offset	0.0%–100.0%	50.0%	These are load cell pre-torque adjustment parameters.
F8-03	Drive gain	0.00–2.00	0.60	
F8-04	Brake gain	0.00–2.00	0.60	

1: Load cell pre-torque compensation

The controller with the load cell sensor pre-outputs the torque matched the load so as to ensure the riding comfort of the elevator.

- Motor driving status: full-load up, no-load down
- Motor braking status: full-load down, no-load up

"Pre-torque offset" is actually the elevator balance coefficient, namely, the percentage of car load to rated load when the car and counterweight are balanced.

"Drive gain" or "Brake gain" scales the elevator's current pre-torque coefficient when the motor runs at the drive or brake side. If the gain set is higher, then the calculated value of startup pre-torque compensation is higher.

The controller identifies the braking or driving state according to the load cell sensor signal, and automatically calculates the required torque compensation value.

When an analog device is used to measure the load, these parameters are used to adjust the elevator startup. To adjust the startup, do as follows:

- In the driving state, increasing the value of F8-03 could reduce the rollback during the elevator startup, but a too high value could cause car lurch at start.
- In braking state, increasing the value of F8-04 could reduce the jerk in command direction during the elevator startup, but a too high value could cause car lurch at start.

2: Automatic pre-torque compensation

This function is enabled only when the ERN1387 or ECN1313 encoder is applied. The system automatically adjusts the compensated torque at startup.

- Gradually increase the value of F2-11 (Zero servo current coefficient) until that rollback is cancelled at brake release and the motor does not vibrate.
- Increase the value of F2-11 (Zero servo current coefficient) if the motor jitters when F2-13 (Zero servo speed loop TI) is less than 1.00.
- Motor vibration and acoustic noise indicate excessive value of F2-12 (Zero servo speed loop KP). Use the default value of F2-12.

■ Mechanical Construction

The mechanical construction affecting the riding comfort involves the guide rail, guide shoe, steel rope, installation of the brake, balance of the car and the resonance caused by the car, guide rail and motor. For asynchronous motor, abrasion or improper installation of the gearbox may arouse bad riding comfort.

1. Installation of the guide rail mainly involves the verticality of the rail, rail surface flatness, smoothness of the guide rail connection and parallelism between two guide rails (including rails at the counterweight side).
2. The tightness of the guide shoes (including the one at the counterweight side) also influences the riding comfort. The guide shoes must not be too loose or tight.
3. The drive from the motor to the car totally depends on the steel rope. Too large flexibility of the steel rope with irregular resistance during the car running may cause curly oscillation of the car. In addition, unbalanced stress of multiple steel ropes may cause the car to jitter during running.
4. The riding comfort may be influenced during running if the brake arm is installed too tightly or the brake is released incompletely.
5. If the car weight is unbalanced, it will cause uneven stress of the guide shoe that connects the car and the guide rail. As a result, the guide shoe will rub with the guide rail during running, affecting the riding comfort.
6. For asynchronous motor, abrasion or improper installation of the gearbox may also affect the riding comfort.
7. Resonance is an inherent character of a physical system, related to the material and quality of system components. If you are sure the oscillation is caused by resonance, reduce the resonance by increasing/decreasing the car weight or counterweight and adding resonance absorbers at connections of the components (for example, place rubber blanket under the motor).

3.3.4 Door Machine Commissioning

As shown in Figure 3-2, correlation of the door machine controller and the elevator controller is that the CTB outputs door open/close command and the door machine controller feeds back the door open/close limit signal.

After commissioning and installation of the door machine are complete, check whether the wiring is correct and the door open/close limit signals are consistent with the default setting. To perform the door machine commissioning, do as follows:

1. In the terminal control mode of the door machine controller, manually short the door open relay output terminal BM/B1 and the door close relay output terminal BM/B2 on the CTB and observe whether the door machine can open and close correspondingly. If the door machine cannot act properly, check whether BM/B1 and BM/B2 are wrongly connected to the input terminals of the door machine controller and whether commissioning of the door machine controller is complete.

2. After ensuring that control of door open/close is normal, check whether the door open/close signal feedback from the door machine is normal.

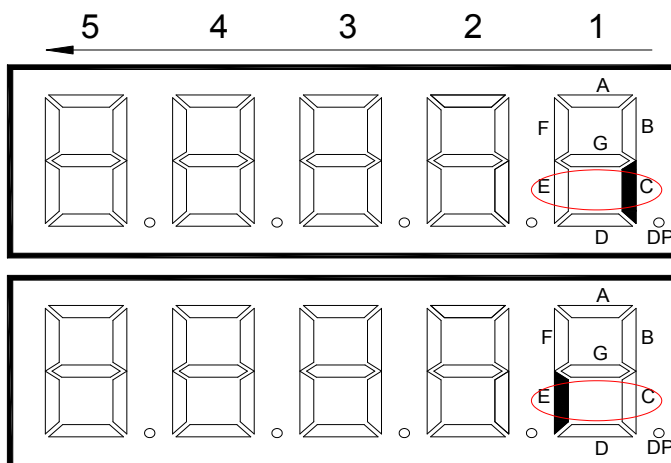
As shown in the following figure which is part of display of parameter F5-35 on the operation panel, segments E and C of the upmost right 7-segment LED are the monitoring points of door open limit and door close limit.

- Segment C ON, segment E OFF: The system receives the door open limit signal and the door is in the open state.
- Segment E ON, segment C OFF: The system receives the door close limit signal and the door is in the close state.

The two segments should be OFF in the door open/close process.

Control the door in the open or close state manually and view the value of F5-35. If the following screen is displayed, it indicates that the door machine controller feeds back the correct door open and close signals.

Figure 3-5 Door open and close limit monitoring signals





4

Function Code Table

Chapter 4 Function Code Table

Function Code	Parameter Name	Setting Range	Default	Unit
Group F0: Basic parameters				
F0-00	Control mode	0: Sensorless flux vector control (SFVC) 1: Closed-loop vector control (CLVC) 2: Voltage/Frequency (V/F) control	1	-
F0-01	Command source selection	0: Operation panel control 1: Distance control	1	-
F0-02	Speed reference in operation panel control	0.050 to F0-04	0.050	m/s
F0-03	Maximum elevator running speed	0.250 to F0-04	1.600	m/s
F0-04	Rated elevator speed	0.250–4.000	1.600	m/s
F0-05	Rated elevator load	300–9999	1000	kg
F0-06	Maximum frequency	20.00–99.00	50.00	Hz
F0-07	Carrier frequency	0.5–16.0	8.0	kHz
Group F1: Motor parameters				
F1-00	Encoder type selection	0: SIN/COS 1: UVW	0	-
F1-01	Rated motor power	0.7–75.0	11.0	kW
F1-02	Rated motor voltage	0–600	380	V
F1-03	Rated motor current	0.00–655.00	21.80	A
F1-04	Rated motor frequency	0.00–99.00	50.00	Hz
F1-05	Rated motor rotational speed	0–3000	147	RPM
F1-06	Encoder initial angle (PMSM)	0.0–359.9	0	Degree
F1-07	Encoder angle at power off (PMSM)	0.0–359.9	0	Degree
F1-08	Wiring mode (PMSM)	0–15	0	-
F1-09	Current filter coefficient (PMSM)	0–40	0	-
F1-10	Encoder check selection	0–65535	0	-
F1-11	Auto-tuning mode selection	0: No operation 1: With-load auto-tuning 2: No-load auto-tuning 3: Shaft auto-tuning	0	-
F1-12	Pulses per revolution of encoder	0–10000	2048	PPR
F1-13	Encoder fault detection time	0–10.0	1.0	s

Function Code	Parameter Name	Setting Range	Default	Unit
F1-14	Stator resistance (asynchronous motor)	0.000–30.000	0	Ω
F1-15	Rotor resistance	0.000–30.000	0	Ω
F1-16	Leakage inductance (asynchronous motor)	0.00–300.00	0	mH
F1-17	Mutual inductance (asynchronous motor)	0.1–3000.0	0	mH
F1-18	Motor magnetizing current (asynchronous motor)	0.01–300.00	0	A
F1-19	Shaft Q inductance (torque)	0.00–650.00	3.00	mH
F1-20	Shaft D inductance (excitation)	0.00–650.00	3.00	mH
F1-21	Back EMF coefficient	0–65535	0	-
F1-25	Motor type	0: Asynchronous motor 1: PMSM	1	-
Group F2: Vector control parameters				
F2-00	Speed loop proportional gain 1	0–100	40	-
F2-01	Speed loop integral time 1	0.01–10.00	0.60	s
F2-02	Switchover frequency 1	0.00–F2-05	2.00	Hz
F2-03	Speed loop proportional gain 2	0–100	35	-
F2-04	Speed loop integral time 2	0.01–10.00	0.80	s
F2-05	Switchover frequency 2	F2-02–F0-05	5.00	Hz
F2-06	Current loop proportional gain	10–500	60	-
F2-07	Current loop integral gain	10–500	30	-
F2-08	Torque limit	0.0–200.0	150.0	%
F2-10	Elevator running direction	0: Running direction and position pulse direction unchanged 1: Running direction reversed, position pulse direction reversed 2: Running direction unchanged, position pulse direction reversed 3: Running direction reversed, position pulse direction unchanged	0	-
F2-11	Zero servo current coefficient	0.20–50.0	15	%
F2-12	Zero servo speed loop KP	0.00–2.00	0.5	-
F2-13	Zero servo speed loop TI	0.00–2.00	0.6	-
F8-01 is set to 2 when the no-load cell startup function is enabled, and the degree of load cell pre-torque compensation is adjusted using F2-11, F2-12, and F2-13.				

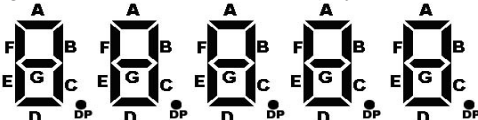
Function Code	Parameter Name	Setting Range	Default	Unit
F2-16	Torque acceleration time	1–500	1	ms
F2-17	Torque deceleration time	1–500	350	ms
These two parameters are used to adjust the time of applying and canceling the current.				
F2-18	Startup acceleration time	0.000–1.500	0.000	s
Group F3: Running control parameters				
F3-00	Startup speed	0.000–0.030	0.000	m/s
F3-01	Startup speed holding time	0.000–0.500	0.000	s
F3-02	Acceleration rate	0.200–2.000	0.600	m/s ²
F3-03	Acceleration start jerk time	0.300–4.000	2.500	s
F3-04	Acceleration end jerk time	0.300–4.000	2.500	s
F3-05	Deceleration rate	0.200–2.000	0.600	m/s ²
F3-06	Deceleration start jerk time	0.300–4.000	2.500	s
F3-07	Deceleration end jerk time	0.300–4.000	2.500	s
F3-08	Special deceleration rate	0.500–2.000	0.900	m/s ²
F3-09	Pre-deceleration distance	0–90.0	0.0	mm
F3-10	Re-leveling speed	0.000–0.080	0.040	m/s
F3-11	Inspection speed	0.100–0.630	0.250	m/s
F3-12	Position of up slow-down 1	0.000–300.00	0.00	m
F3-13	Position of down slow-down 1	0.000–300.00	0.00	m
F3-14	Position of up slow-down 2	0.000–300.00	0.00	m
F3-15	Position of down slow-down 2	0.000–300.00	0.00	m
F3-16	Position of up slow-down 3	0.00–300.00	0.00	m
F3-17	Position of down slow-down 3	0.000–300.00	0.00	m
F3-18	Zero-speed control time at startup	0.000–1.000	0.200	s
F3-19	Brake release delay	0.000–1.000	0.600	s
F3-20	Zero-speed control time at end	0.200–1.500	0.200	s
F3-21	Low-speed re-leveling speed	0.080 to F3-11	0.100	m/s
Group F4: Floor parameters				
F4-00	Leveling adjustment	0–60	30	mm
F4-01	Current floor	F6-01 to F6-00	1	Number of pulses
F4-02	High byte of current floor position	0–65535	1	Number of pulses
F4-03	Low byte of current floor position	0–65535	34464	Number of pulses

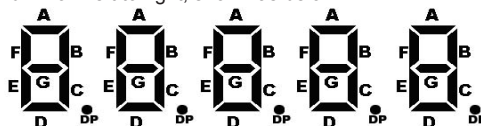
Function Code	Parameter Name	Setting Range	Default	Unit
F4-04	Length 1 of leveling flag	0-65535	0	Number of pulses
F4-05	Length 2 of leveling flag	0-65535	0	Number of pulses
F4-06	High byte of floor 1 height	0-65535	0	Number of pulses
F4-07	Low byte of floor 1 height	0-65535	0	Number of pulses
F4-08	High byte of floor 2 height	0-65535	0	Number of pulses
F4-09	Low byte of floor 2 height	0-65535	0	Number of pulses
F4-10	High byte of floor 3 height	0-65535	0	Number of pulses
F4-11	Low byte of floor 3 height	0-65535	0	Number of pulses
F4-12	High byte of floor 4 height	0-65535	0	Number of pulses
F4-13	Low byte of floor 4 height	0-65535	0	Number of pulses
F4-14	High byte of floor 5 height	0-65535	0	Number of pulses
F4-15	Low byte of floor 5 height	0-65535	0	Number of pulses
F4-16	High byte of floor 6 height	0-65535	0	Number of pulses
F4-17	Low byte of floor 6 height	0-65535	0	Number of pulses
F4-18	High byte of floor 7 height	0-65535	0	Number of pulses
F4-19	Low byte of floor 7 height	0-65535	0	Number of pulses
F4-20	High byte of floor 8 height	0-65535	0	Number of pulses
F4-21	Low byte of floor 8 height	0-65535	0	Number of pulses
F4-22	High byte of floor 9 height	0-65535	0	Number of pulses
F4-23	Low byte of floor 9 height	0-65535	0	Number of pulses
	...			

Function Code	Parameter Name	Setting Range	Default	Unit
F4-82	High byte of floor 39 height	0–65535	0	Number of pulses
F4-83	Low byte of floor 39 height	0–65535	0	Number of pulses
Group F5: Terminal function parameters				
F5-00	Attendant/Normal switchover time	3–200	3	s
F5-01	X1 function selection	NO/NC input: 00: No function 01/33: Up leveling signal 02/34: Down leveling signal 03/35: Door zone signal 04/36: Safety circuit feedback signal 05/37: Door lock circuit feedback signal 06/38: Main contactor feedback signal 07/39: Brake feedback signal 08/40: Inspection signal 09/41: Inspection up signal 10/42: Inspection down signal 11/43: Fire emergency signal 12/44: Up limit signal 13/45: Down limit signal 14/46: Overload signal 15/47: Full-load signal 16/48: Up slow-down 1 signal 17/49: Down slow-down 1 signal 18/50: Up slow-down 2 signal 19/51: Down slow-down 2 signal (To be continued)	1	-
F5-02	X2 function selection		0	-
F5-03	X3 function selection		2	-
F5-04	X4 function selection		0	-
F5-05	X5 function selection		5	-
F5-06	X6 function selection		6	-
F5-07	X7 function selection		7	-
F5-08	X8 function selection		59	-
F5-09	X9 function selection		40	-
F5-10	X10 function selection	(To be continued)	9	-

Function Code	Parameter Name	Setting Range	Default	Unit
F5-11	X11 function selection	20/52: Up slow-down 3 signal	10	-
F5-12	X12 function selection	21/53: Down slow-down 3 signal	44	-
F5-13	X13 function selection	22/54: Shorting door lock circuit contactor feedback	45	-
F5-14	X14 function selection	23/55: Firefighter running signal	48	-
F5-15	X15 function selection	24/56: Door machine 1 light curtain signal	49	-
F5-16	X16 function selection	25/57: Door machine 2 light curtain signal	0	-
F5-17	X17 function selection	26/58: Brake feedback 2 signal	0	-
F5-18	X18 function selection	27/59: UPS valid signal	43	-
F5-19	X19 function selection	28/60: Elevator locked signal	0	-
F5-20	X20 function selection	29/61: Safety circuit 2 feedback signal	0	-
F5-21	X21 function selection	30/62: Shorting PMSM stator feedback signal	30	-
F5-22	X22 function selection	31/63: Door lock circuit 2 feedback signal	0	-
F5-23	X23 function selection	32/64: Reserved	47	-
F5-24	X24 function selection	65/97: Door machine 1 safety edge signal 66/98: Door machine 2 safety edge signal 67/99: Motor overheat signal 68/100: Earthquake signal 69/101: Back door forbidden signal 70/102: Light-load signal 71/103: Half-load signal 72/104: Fire emergency base station switchover signal 73/105: False floor signal (End)	46	-

Function Code	Parameter Name	Setting Range	Default	Unit	
F5-25	CTB input type selection		320	-	
Note: By default, Bit8 is used as the light-load signal NO input to support the CTB with nine inputs. In this case, the default value of F5-25 is 320.					
Bit	Parameter Name	Default	Bit	Parameter Name	Default
Bit0	Door 1 light curtain	0	Bit5	Door 2 close limit	0
Bit1	Door 2 light curtain	0	Bit6	Full-load signal (digital)	1
Bit2	Door 1 open limit	0	Bit7	Overload signal (digital)	0
Bit3	Door 2 open limit	0	Bit8	Light-load signal (digital)	1
Bit4	Door 1 close limit	0	0: NO 1: NC		
F5-26	Y1 function selection	Output function: 0: No function 1: RUN contactor control 2: Brake contactor control 3: Shorting door lock circuit contactor 4: Fire emergency floor arrival signal feedback 5: Door 1 open 6: Door 1 close 7: Door 2 open 8: Door 2 close 9: Brake and RUN contactors healthy 10: Fault state 11: Running monitor 12: Shorting PMSM stator contactor 13: Emergency evacuation automatic switchover 14: System healthy 15: Emergency buzzer control 16: High-voltage startup of brake 17: Elevator running in up direction 18: Fan running 19: Medical sterilization 20: Non-door zone stop 21: Electric lock 22: Non-service state		1	-
F5-27	Y2 function selection			2	-
F5-28	Y3 function selection			12	-
F5-29	Y4 function selection			0	-
F5-30	Y5 function selection			0	-
F5-31	Y6 function selection			13	-

Function Code	Parameter Name	Setting Range	Default	Unit
F5-32	Communication status display		0	-
Figure 4-1 Communication status display  The upmost left 7-segment LED is expressed by 0–9, and used for CAN communication monitoring. • 0: communication normal • 9: communication interrupted completely The eight segments A/B/C/D/E/F/G/DP of the four 7-segment LEDs from right to left successively indicate Modbus communication monitoring of each HOP. • ON: The Modbus communication of the HOP for the floor is normal. • OFF: The Modbus communication of the HOP for the floor is interrupted.				
F5-33	Program control selection	Bit0 to Bit3: Reserved Bit3: Elevator fire emergency requirement for Hongkong Bit4: Arrival gong disabled at night Bit6: Door lock disconnected in the prerequisite of inspection switched over to normal running Bit7: Fault code not displayed on the keypad Bit8: Door open command cancelled immediately at door open limit Bit9: Car stop due to zero-speed torque holding when the brake feedback is abnormal	0	-
F5-34	Terminal state display		0	-
F5-35	Terminal state display		0	-

Function Code	Parameter Name		Setting Range	Default	Unit
F5-34/F5-35 indicates the state of the I/O terminals. The serial numbers of the five 7-segment LEDs are 5, 4, 3, 2 and 1 from left to right, shown as below:					
					
F5-34			F5-35		
SN	Segment	Indication	SN	Segment	Indication
1			1	A	Light curtain 1
	B	Up leveling signal		B	Light curtain 2
	C	Down leveling signal		C	Door 1 open limit
	D	Door zone signal		D	Door 2 open limit
	E	Safety circuit feedback 1		E	Door 1 close limit
	F	Door lock circuit feedback 1		F	Door 2 close limit
	G	RUN contactor feedback		G	Full-load signal
	DP	Brake contactor feedback 1		DP	Overload signal
2	A	Inspection signal	2	A	Door open button
	B	Inspection up signal		B	Door close button
	C	Inspection down signal		C	Door open delay button
	D	Fire emergency signal		D	Bypass signal
	E	Up limit signal		E	Attendant signal
	F	Down limit signal		F	Direction change signal
	G	Overload signal		G	Independent running signal
	DP	Full-load signal		DP	Firefighter operation signal
3	A	Up slow-down 1 signal	3	A	Door open output 1
	B	Down slow-down 1 signal		B	Door close output 1
	C	Up slow-down 2 signal		C	Door lock signal
	D	Down slow-down 2 signal		D	Door open output 1
	E	Up slow-down 3 signal		E	Door close output 2
	F	Down slow-down 3 signal		F	Door lock signal
	G	Shorting door lock circuit contactor feedback		G	Up arrival gong
	DP	Motor overheat signal		DP	Down arrival gong

Function Code		Parameter Name	Setting Range		Default	Unit
4	A	Door machine 1 light curtain	4	A	Door open button display	
	B	Door machine 2 light curtain		B	Door close button display	
	C	Brake feedback 2		C	Door open delay button display	
	D	UPS input		D	Bypass sign	
	E	Elevator locked input		E	Reserved	
	F	Safety circuit feedback 2		F	Buzzer output	
	G	Shorting PMSM stator contactor feedback		G	Reserved	
	DP	Door lock circuit feedback 2		DP	Energy saving sign	
5	A	Reserved	5	A	System light curtain state 1	
	B	RUN contactor output		B	System light curtain state 2	
	C	Brake contactor output		C	Hall call elevator-locked input	
	D	Shorting door lock circuit contactor control		D	Hall call fire emergency input	
	E	Fire emergency floor arrival signal		E	Full-load signal	
				F	Over-load signal	
F5-36	Load cell input selection	0: Invalid 1: CTB digital input 2: CTB analog input 3: MCB analog input		1	-	
F5-37	X25 function selection	0: No function 4: Safety circuit signal 5: Door lock circuit signal (High-voltage detection points, these functions can be repeatedly selected for function codes)		4	-	
F5-38	X26 function selection			5	-	
F5-39	X27 function selection			5	-	
Group F6: Basic elevator parameters						
F6-00	Top floor of the elevator	F6-01 to 40		09	-	
F6-01	Bottom floor of the elevator	1 to F6-00		1	-	
F6-02	Fire emergency floor	F6-01 to F6-00		1	-	
F6-03	Elevator-locked floor	F6-01 to F6-00		1	-	
F6-04	Service floors 1	F6-01 to F6-00		1	-	
F6-05	Fire emergency floor	0-65535		65535	-	
F6-06	Service floors 2	0-65535		65535	-	
F6-07	Number of elevators in group mode	1-8		1	-	
F6-08	Elevator SN	1-8		1	-	

Function Code	Parameter Name	Setting Range	Default	Unit
F6-09	Program selection	Bit0: Dispersed waiting Bit1: Reserved Bit2: Parallel implemented at monitoring port Bit3: Parallel control with CAN2 Bit4: Group control with CTRL80(M) Bit6: Clear floor number in advance and display direction Bit8: Hall call direction not displayed Bit10: E30 cleared at re-leveling Bit14: Detection of safety circuit 2 and door lock circuit 2 at intervals	0	-
Note: The CTRL80(M2) supports parallel control with CAN1/CAN2/monitoring port. By default, CAN1 is used for parallel control. The function of not displaying the hall call direction is used only for scenarios in which only one hall call button is available.				
F6-10	Leveling sensor filter time	10–50	14	ms
F6-11	Elevator function selection	Bit1: Canceling check at returning to base floor Bit2: Cancelling auto sequential arrange of floor addresses to be displayed Bit5: Enabling current detection at startup for PMSM Bit6: Reversing MCB light output Bit7: Door open valid at non-door zone in the inspection state Bit8: Door open and close once after inspection turned to normal Bit10: Buzzer not tweet upon re-leveling Bit11: Super short floor function Bit13: E53 fault auto reset Bit14: Super short floor up slow-down not reset Bit15: Super short floor down slow-down not reset	256	-
F6-12	VIP floor	0 to F6-00	0	-
F6-13	Security floor	0 to F6-00	0	-
F6-14	Down-collective 1 start time	00.00–23.59	0	-
F6-15	Down-collective 1 end time	00.00–23.59	0	-

Function Code	Parameter Name	Setting Range	Default	Unit
F6-16	Down-collective 2 start time	00.00–23.59	0	-
F6-17	Down-collective 2 end time	00.00–23.59	0	-
F6-18	Time-based floor service 1 start time	00.00–23.59	0	-
F6-19	Time-based floor service 1 end time	00.00–23.59	0	-
F6-20	Floor 1 of time-based floor service 1	0–65535	65535	-
F6-21	Floor 2 of time-based floor service 1	0–65535	65535	-
F6-22	Time-based floor service 2 start time	00.00–23.59	0	-
F6-23	Time-based floor service 2 end time	00.00–23.59	0	-
F6-24	Floor 1 of time-based floor service 2	0–65535	65535	-
F6-25	Floor 2 of time-based floor service 2	0–65535	65535	-
F6-26	Peak 1 start time	00.00–23.59	0	-
F6-27	Peak 1 end time	00.00–23.59	0	-
F6-28	Peak 1 floor	F6-01 to F6-00	1	-
F6-29	Peak 2 start time	00.00–23.59	0	-
F6-30	Peak 2 end time	00.00–23.59	0	-
F6-31	Peak 2 floor	F6-01 to F6-00	1	-
F6-35	Floor 3	0–65535	65535	-
F6-36	Floor 3 of time-based floor service 1	0–65535	65535	-
F6-37	Floor 3 of time-based floor service 2	0–65535	65535	-
F6-38	Elevator lock start time	00.00–23.59	0	-
F6-39	Elevator lock end time	00.00–23.59	0	-

Function Code	Parameter Name	Setting Range	Default	Unit
F6-40	Program control selection 1	Bit0: Disability function Bit1: Soft limit function Bit2: JP16 input used as the back door selection Bit3: JP16 input used as the back door open signal Bit4: Only one door of opposite doors open under manual control Bit5: Timed elevator lock Bit6: Manual door Bit7: Elevator lock/Fire emergency under hall call at any floor Bit9: Disabling reverse floor number clear Bit10: Next arriving floor number displayed Bit11: Car call responded first Bit12: Car call assisted command in single door used as disability function Bit13: Folding command used as disability function when Bit13 is 1 and back door function when Bit13 is 0 Bit14: Car call command folding Bit15: JP20 used for switchover to back door	0	-

Note:

For details on the function for the disabled people and opposite door function, see Chapter 5.

The soft limit function is described as follows:

When the up slow-down 1 and down leveling signal are valid and the up leveling signal is invalid, the system considers that the up limit signal is valid. It is the same for the down limit signal.

Function Code	Parameter Name	Setting Range	Default	Unit
F6-41	Program control selection 2	Bit2: Inspection to stop due to slow-down 1 Bit4: Buzzer tweet during door open delay Bit6: Repeat command cancelled during door open delay Bit8: Elevator lock at door open Bit9: Display available at elevator lock Bit10: Elevator lock in the attendant state Bit11: Blinking at arrival (within the time set in F6-47) Bit12: Door re-open during door open delay Bit13: Door re-open after car call of the present floor	0	-
F6-42	Program control selection 3	Bit1: Door open/close command cancelled at delay after door open/close limit Bit2: Door lock state not judged at door close output Bit3: Door close command output during running	0	-
F6-43	Attendant function selection	Bit0: Call cancelled in attendant state Bit1: Not responding to hall call Bit2: Automatic exiting the attendant state Bit3: Door close at jogging Bit4: Automatic door close Bit5: Buzzer tweet at intervals in attendant state Bit6: Continuous buzzer tweet in attendant state	34	-

Function Code	Parameter Name	Setting Range	Default	Unit
F6-44	Fire emergency function selection	Bit3: Arrival gong output in fire emergency state Bit4: Multiple car calls registered in fire emergency state Bit5: Retentive at power failure in fire emergency state Bit6: Door close by holding down the door close button Bit7: Door close at low speed Bit8: Door close during car call register Bit9: Hall call displayed in fire emergency state Bit11: Exit fire emergency state for firefighter Bit14: Door open by holding down the door open button Bit15: Automatic door open in fire emergency floor	19312	-
F6-45	Emergency evacuation function selection	0–65535	45056	-
F6-46	VIP function selection	Bit0: VIP hall call enabled (for VIP floor) Bit1: Terminal for VIP hall call enabled Bit8: Number of VIP car calls limited	0	s
F6-47	Blinking advance time	0.0–15.0	0	s
F6-48	Emergency evacuation switching speed	0.010–0.630	0.010	-
F6-49	Evacuation parking floor	0 to F6-01	0	-
Group F7: Test function parameters				
F7-00	Car call floor registered	0 to F6-00	0	-
F7-01	Up call floor registered	0 to F6-00	0	-
F7-02	Down call floor registered	0 to F6-00	0	-
F7-03	Random test times	0–60000	0	-
F7-04	Hall call enabled	0: Hall call allowed 1: Hall call prohibited	0	-
F7-05	Door open enabled	0: Door open allowed 1: Door open prohibited	0	-
F7-06	Over-load function selection	0: Over-load running prohibited 1: Over-load running allowed	0	-

Function Code	Parameter Name	Setting Range	Default	Unit
F7-07	Limit switch	0: Limit switch enabled 1: Limit switch disabled	0	-
F7-08	Random running interval	0–1000	0	s
Group F8: Enhanced function parameters				
F8-00	Load cell auto-tuning	0–100	0	%
F8-01	Pre-torque selection	0: Pre-torque invalid 1: Load cell pre-torque compensation 2: Automatic pre-torque compensation	0	-
F8-02	Pre-torque offset	0.0–100.0%	50.0	%
F8-03	Drive gain	0.00–2.00	0.60	-
F8-04	Brake gain	0.00–2.00	0.60	-
F8-06	Car no-load	0–1023	0	-
F8-07	Car full-load	0–1023	100	-
F8-08	Anti- nuisance function	Bit0: Load cell judging (load cell sensor or switch required) Bit1: Light curtain judging Bit2: Light-load judging	0	-
F8-09	Emergency evacuation operation speed at power failure	0.000–0.100	0.050	m/s
F8-10	Emergency evacuation operation mode at power failure	0: Motor not running 1: UPS 2: 48 V battery power supply	1	-
F8-11	Brake apply delay	0.000–1.000	0.300	s
F8-12	Fire-emergency floor 2	0 to F6-00	0	-
F8-14	Hall call communication setting	Bit4: Hall call energy-saving control	0	-
F8-16	Start address of hall call assisted command	0–40	0	-
F8-17	Hall call address check	0–1	0	-
Note: When F8-17 is set to 1, HOPs at all floors display the set addresses for installation check. When F8-17 is set to 0, the normal display is restored.				
Group F9: Time parameters				
F9-00	Time for idle elevator to return to base floor	0–240	10	min
F9-01	Time for fan and lamp to be turned off	0–240	2	min
F9-02	Maximum floor running time interval	20–45	45	s

Function Code	Parameter Name	Setting Range	Default	Unit
F9-03	Clock: year	2000–2100	2011	YY
F9-04	Clock: month	1–12	1	MM
F9-05	Clock: day	1–31	1	DD
F9-06	Clock: hour	0–23	0	HH
F9-07	Clock: minute	0–59	0	MM
F9-09	Accumulative running time	0–65535	0	h
F9-11	High byte of running times	0–9999	0	-
F9-12	Low byte of running times	0–9999	0	-
Group FA: Keypad setting parameters				
FA-00	Keypad display selection	0: Reversed display 1: Positive display 2: Reversed display, hall call data 3: Positive display, hall call data	3	-
FA-01	Run display	1–65535	65535	-
FA-02	Stop display	1–65535	65535	-
FA-03	Current encoder angle	0.0–359.9	0.0	°
FA-04	Reserved	-	-	-
FA-05	Software (ZK)	0–65535	0	-
FA-06	Software (DSP)	0–65535	0	-
FA-07	Heatsink temperature	0–100	0	°C
FA-11	Pre-torque current	0.0–200.0	1	%
FA-12	Logic information	0–65535	0	-
FA-13	Curve information	0–65535	0	-
FA-14	Set speed	0.000–4.000	0	m/s
FA-15	Feedback speed	0.000–4.000	0	m/s
FA-16	Bus voltage	0–999.9	0	V
FA-17	Present position	0.0–300.0	0	m
FA-18	Output current	0.0–999.9	0	A
FA-19	Output frequency	0.00–99.99	0	Hz
FA-20	Torque current	0.0–999.9	0	A
FA-21	Output voltage	0–999.9	0	V
FA-22	Output torque	0–100	0	%
FA-23	Output power	0.00–99.99	0	kW
FA-24	Communication interference	0–65535	0	-
FA-25	Encoder interference	0–65535	0	-

Function Code	Parameter Name	Setting Range	Default	Unit
FA-26	Input state 1	0-65535	0	-
FA-27	Input state 2	0-65535	0	-
FA-28	Input state 3	0-65535	0	-
FA-29	Input state 4	0-65535	0	-
FA-31	Output state 1	0-65535	0	-
FA-32	Output state 2	0-65535	0	-
FA-33	Car input state	0-65535	0	-
FA-34	Car output state	0-65535	0	-
FA-35	Hall state	0-65535	0	-
FA-36	System state 1	0-65535	0	-
FA-37	System state 2	0-65535	0	-
FA-46	Hall call communication state 1	0-65535 (floors 1 to 16)	0	-
FA-47	Hall call communication state 2	0-65535 (floors 17 to 32)	0	-
FA-48	Hall call communication state 3	0-65535 (floors 33 to 40)	0	-
Group FB: Door function parameters				
FB-00	Number of door machine(s)	1-2	1	-
FB-01	CTB software	00-999	0	-
FB-02	Service floor 1 of door machine 1	0-65535	65535	-
FB-03	Service floor 2 of door machine 1	0-65535	65535	-
FB-04	Service floor 1 of door machine 2	0-65535	65535	-
FB-05	Service floor 2 of door machine 2	0-65535	65535	-
FB-06	Open door time protection	5-99	10	s
FB-07	Arrival gong output delay	0-1000	0	ms
FB-08	Close door time protection	5-99	15	s
FB-09	Times of door open and close	0-20	5	-
FB-10	Door state of idle elevator	0: Close the door successfully 1: Waiting with door open at base floor 2: Waiting with door open at each floor	0	-
FB-11	Door open holding time for hall call	1-1000	5	s

Function Code	Parameter Name	Setting Range	Default	Unit
FB-12	Door open holding time for car call	1–1000	3	s
FB-13	Door open holding time at base floor	1–1000	10	s
FB-14	Door open holding time	10–1000	30	s
FB-15	Special door open holding time	10–1000	30	s
FB-16	Manual holding time	1–60	5	s
Note: FB-15 (Special door open holding time) specifies the door open holding time for disabled people. FB-16 (Manual holding time) specifies the time for the system to determine door close limit logically when you open/close the door with hands.				
FB-17	Forced door close time	5–180	120	s
FB-18	Service floor 3 of door machine 1	0–65535	65535	-
FB-19	Service floor 3 of door machine 2	0–65535	65535	-
Group FC: Protection function parameters				
FC-00	Program control selection	Bit0: Detection of short circuit to ground at power-on Bit2: Deceleration to stop at light curtain valid Bit9: Without door open/close limit	0	-
FC-01	Protection function selection	Bit0: Over-load protection Bit1: Output phase loss Bit2: Over-modulation function Bit4: Light curtain judgment at door close limit Bit5: DSP communication check Bit6: MCB over-speed judgment Bit7: Stop at leveling position function Bit14: Input phase loss	65	-
FC-02	Over-load protection coefficient	0.50–10.00	1.00	-
FC-03	Over-load alarm coefficient	50–100	80	%
FC-04	Opposite door selection	0–3	0	-
FC-06	Specified fault setting	0–99	0	-
FC-07	Specified fault	0–9999	0	-
FC-08	Specified fault subcode	0–65535	0	-
FC-09	Month and day of specified fault	0–1231	0	MM.DD

Function Code	Parameter Name	Setting Range	Default	Unit
FC-10	Time of specified fault	0–23.59	0	HH.MM
FC-11	Logic information of specified fault	0–65535	0	-
FC-12	Curve information of specified fault	0–65535	0	-
FC-13	Set speed upon specified fault	0.000–4.000	0	m/s
FC-14	Feedback speed upon specified fault	0.000–4.000	0	m/s
FC-15	Bus voltage upon specified fault	0–999.9	0	V
FC-16	Present position upon specified fault	0.0–300.0	0	m
FC-17	Output current upon specified fault	0.0–999.9	0	A
FC-18	Output frequency upon specified fault	0.00–99.99	0	Hz
FC-19	Torque current upon specified fault	0.0–999.9	0	A
FC-20	1st fault	0–9999	0	-
FC-21	1st fault subcode	0–65535	0	-
FC-22	Month and day of 1st fault	0–1231	0	MM.DD
FC-23	Time of 1st fault	0–23.59	0	HH.MM
FC-24	2nd fault	0–9999	0	-
FC-25	2nd fault subcode	0–65535	0	-
FC-26	Month and day of 2nd fault	0–1231	0	MM.DD
FC-27	Time of 2nd fault	0–23.59	0	HH.MM
FC-28	3rd fault	0–9999	0	-
FC-29	3rd fault subcode	0–65535	0	-
FC-30	Month and day of 3rd fault	0–1231	0	MM.DD
FC-31	Time of 3rd fault	0–23.59	0	HH.MM
FC-32	4th fault	0–9999	0	-
FC-33	4th fault subcode	0–65535	0	-
FC-34	Month and day of 4th fault	0–1231	0	MM.DD
FC-35	Time of 4th fault	0–23.59	0	HH.MM
	...			-
FC-56	10th fault	0–9999	0	-
FC-57	10th fault subcode	0–65535	0	-
FC-58	Month and day of 10th fault	0–1231	0	MM.DD

Function Code	Parameter Name	Setting Range	Default	Unit
FC-59	Time of 10th fault	0–23.59	0	HH.MM
FC-60	Latest fault	0–9999	0	-
FC-61	Latest fault subcode	0–65535	0	-
FC-62	Month and day of latest fault	0–1231	0	MM.DD
FC-63	Time of latest fault	0–23.59	0	HH.MM
FC-64	Logic information of latest fault	0–65535	0	-
FC-65	Curve information of latest fault	0–65535	0	-
FC-66	Set speed upon latest fault	0.000–4.000	0	m/s
FC-67	Feedback speed upon latest fault	0.000–4.000	0	m/s
FC-68	Bus voltage upon latest fault	0–999.9	0	V
FC-69	Preset position upon latest fault	0.0–300.0	0	m
FC-70	Output current upon latest fault	0.0–999.9	0	A
FC-71	Output frequency upon latest fault	0.00–99.99	0	Hz
FC-72	Torque current upon latest fault	0.0–999.9	0	A
Group FD: Communication parameters				
FD-00	Baud rate setting	0: 9600 1: 38400	0	bit/s
FD-02	Local address	0–127	1	-
FD-03	Response delay	0–20	10	ms
FD-04	Communication timeout delay	0–60.0	0.0	s
FD-05	Stop delay upon re-leveling	0.00–2.00	0.00	s
FD-07	HCB-JP1 input selection	0: Reserved 1: Elevator lock signal 2: Fire emergency signal 3: Present floor forbidden 4: VIP floor signal 5: Security floor signal 6: Door close button input (button)	1	-
FD-08	HCB-JP2 input selection		2	-

Function Code	Parameter Name	Setting Range	Default	Unit
FD-09	HCB-JP1 output selection	0: Reserved 1: Up arrival gong 2: Down arrival gong 3: Fault signal	1	-
FD-10	HCB-JP2 output selection	4: Non-door zone stop 5: Non-service state output 6: Door close button output (button)	2	-
FD-11	HCB-B-JP1 input selection	0: Reserved 1: Light-load signal 2: Light-load signal 3: Door 2 selection (door 1 or 2 selection) 4: Door 2 restriction (forbidding operation of back door) 5: Door 1 safety edge 6: Door 2 safety edge 7: Single/Double door selection	0	-
FD-12	HCB-B-JP2 input selection		0	-
FD-13	HCB-B-JP3 input selection		0	-
FD-14	HCB-B-JP4 input selection		0	-
FD-15	HCB-B-JP5 input selection		0	-
FD-16	HCB-B-JP6 input selection		0	-
FD-17	HCB-B-A1 output selection	0: Reserved 1: Fault output 2: Non-door zone stop 3: Non-service state output 4: Fire emergency output 5: Emergency evacuation output at power failure 6: Door lock valid 7: Night output	0	-
FD-18	HCB-B-A2 output selection		0	-
FD-19	HCB-B-B1 output selection		0	-
FD-20	HCB-B-B2 output selection		0	-
FD-21	HCB-B-C1 output selection		0	-
FD-22	HCB-B-C2 output selection		0	-
FD-23	HCB-B-C3 output selection		0	-
FD-24	HCB-B-C4 output selection		0	-
FD-25	HCB-B-C5 output selection		0	-
FD-26	HCB-B-C6 output selection		0	-
Group FE: Elevator function parameters				
FE-00	Collective selective mode	0: Full selective 1: Down selective 2: Up selective	0	-
FE-01	Floor 1 corresponding display	The first two digits indicate the display code of ten's place, and the last two digits indicate the display code of unit' s place. 00: display 0 01: display 1 (To be continued)	1901	-
FE-02	Floor 2 corresponding display		1902	-
FE-03	Floor 3 corresponding display		1903	-
FE-04	Floor 4 corresponding display		1904	-

Function Code	Parameter Name	Setting Range	Default	Unit
FE-05	Floor 5 corresponding display	02: display 2 03: display 3 04: display 4 05: display 5 06: display 6 07: display 7 08: display 8 09: display 9 10: display A 11: display B 12: display G 13: display H 14: display L 15: display M 16: display P 17: display R 18: display - 19: no display 20: display 12 21: display 13 (To be continued)	1905	-
FE-06	Floor 6 corresponding display		1906	-
FE-07	Floor 7 corresponding display		1907	-
FE-08	Floor 8 corresponding display		1908	-
FE-09	Floor 9 corresponding display		1909	-
FE-10	Floor 10 corresponding display		0100	-
FE-11	Floor 11 corresponding display		0101	-
FE-12	Floor 12 corresponding display		0102	-
FE-13	Floor 13 corresponding display		0103	-
FE-14	Floor 14 corresponding display		0104	-
FE-15	Floor 15 corresponding display		0105	-
FE-16	Floor 16 corresponding display		0106	-
FE-17	Floor 17 corresponding display		0107	-
FE-18	Floor 18 corresponding display		0108	-
FE-19	Floor 19 corresponding display		0109	-
FE-20	Floor 20 corresponding display		0200	-
FE-21	Floor 21 corresponding display		0201	-
FE-22	Floor 22 corresponding display		0202	-
FE-23	Floor 23 corresponding display		0203	-
FE-24	Floor 24 corresponding display		0204	-
FE-25	Floor 25 corresponding display		0205	-

Function Code	Parameter Name	Setting Range	Default	Unit
FE-26	Floor 26 corresponding display	22: display 23 23: display C 24: display D 25: display E 26: display F 27: display I 28: display J 29: display K 30: display N 31: display O 32: display Q 33: display S 34: display T 35: display U 36: display V 37: display W 38: display X 39: display Y 40: display Z (End)	0206	-
FE-27	Floor 27 corresponding display		0207	-
FE-28	Floor 28 corresponding display		0208	-
FE-29	Floor 29 corresponding display		0209	-
FE-30	Floor 30 corresponding display		0300	-
FE-31	Floor 31 corresponding display		0301	-
FE-35	Floor 32 corresponding display		0	-
FE-36	Floor 33 corresponding display		0	-
FE-37	Floor 34 corresponding display		0	-
FE-38	Floor 35 corresponding display		0	-
FE-39	Floor 36 corresponding display		0	-
FE-40	Floor 37 corresponding display		0	-
FE-41	Floor 38 corresponding display		0	-
FE-42	Floor 39 corresponding display		0	-
FE-43	Floor 40 corresponding display		0	-
FE-52	Display setting 1 of highest digit		0	-
FE-53	Display setting 2 of highest digit		0	-
FE-54	Display setting 3 of highest digit		0	-
FE-55	Display setting 4 of highest digit		0	-
FE-56	Display setting 5 of highest digit		0	-
The display setting of highest digit is used as follows: When the two-digit floor display cannot meet the requirements, you can add the third digit display by using "display setting of highest digit". In any value of FE-52 to FE-56, the two high bits indicate the physical address of the floor, and the two low bits indicate the display content. For example, if you require that the display of floor 18 is 17A, set FE-18 to 0710 (expressing "7A), and then set the highest digit display, for example, setting FE-52 to 1801 (indicating that the highest digit display for the floor with address 18 is "1").				

Function Code	Parameter Name	Setting Range	Default	Unit
FE-32	Elevator function selection 1	Bit2: Re-leveling function Bit3: Pre-open function Bit4: Removal of stuck hall calls Bit5: Night security floor function Bit6: Down collective selective peak service Bit7: Parallel (group mode) peak service Bit8: Floor selection of time-based floor service Bit9: VIP floor function Bit11: Mal-deletion of car calls Bit12: Mal-deletion of hall calls Bit15: Shorting door lock circuit detection function valid	34816	-
FE-33	Elevator function selection 2	Bit1: Door open holding at door open limit Bit2: Not output door close signal at door close limit Bit4: Auto reset for RUN and brake contactor stuck Bit5: Detecting slow-down switch stuck Bit7: Forced door close function Bit8: NC output of shorting motor stator contactor Bit9: Immediately stop upon re-leveling Bit13: High-speed elevator protection function Bit15: Opposite door independent control	36	-
FE-34	Factory contract number	0–9999	0	-
Group FF: Factory parameters				
Group FP: User parameters				
FP-00	User password	0: no password	0	-
FP-01	Parameter update	0: No update 1: Restore factory setting 2: Clear retentive information	0	-
FP-02	User setting check	0: Disabled 1: Enabled	0	-
Group A1: Load cell parameters				
A1-00	Light-load setting	2%–50%	20	%

Function Code	Parameter Name	Setting Range	Default	Unit
A1-01	Full-load setting	30%–100%	80	%
A1-02	Overload setting	80%–120%	110	%
A1-03	Setting of car full-load at floors 1 and 2	0–65535	0	-
A1-04	Setting of car full-load at floors 3 and 4	0–65535	0	-
A1-05	Setting of car full-load at floors 5 and 6	0–65535	0	-
A1-06	Setting of car full-load at floors 7 and 8	0–65535	0	-
A1-07	Setting of car full-load at floors 9 and 10	0–65535	0	-
A1-08	Setting of car full-load at floors 11 and 12	0–65535	0	-
A1-09	Setting of car full-load at floors 13 and 14	0–65535	0	-
A1-10	Setting of car full-load at floors 15 and 16	0–65535	0	-
A1-11	Setting of car full-load at floors 17 and 18	0–65535	0	-
A1-12	Setting of car full-load at floors 19 and 20	0–65535	0	-
A1-13	Setting of car full-load at floors 21 and 22	0–65535	0	-
A1-14	Setting of car full-load at floors 23 and 24	0–65535	0	-
A1-15	Setting of car full-load at floors 25 and 26	0–65535	0	-
A1-16	Setting of car full-load at floors 27 and 28	0–65535	0	-
A1-17	Setting of car full-load at floors 29 and 30	0–65535	0	-
A1-18	Setting of car full-load at floors 31 and 32	0–65535	0	-
A1-19	Setting of car full-load at floors 33 and 34	0–65535	0	-
A1-20	Setting of car full-load at floors 35 and 36	0–65535	0	-
A1-21	Setting of car full-load at floors 37 and 38	0–65535	0	-
A1-22	Setting of car full-load at floors 39 and 40	0–65535	0	-



5

System Application

Chapter 5 System Application

5.1 Parallel Mode and Group Mode

5.1.1 Parallel Control of Two Elevators

Compatible with the CTRL80(M), the CTRL80(M2) supports parallel control of two elevators, and group control of a maximum of eight elevators if the group control board is used.

The CTRL80(M2) provides two CAN communication ports CAN1 (CN3), CAN2 (CN4) and an RS232 (CN5) communication port. All the three communication ports can be used for parallel control. The following table lists the parameter setting of parallel control.

Table 5-1 Parameter setting of parallel control

Function Code	Parameter Name	Setting Range	Setting in Parallel
F6-07	Number of elevators in group mode	1–8	2
F6-08	Elevator SN	1–8	Master elevator: 1
			Slave elevator: 2
F6-09	Program selection	Bit2: Parallel implemented at monitoring port	Bit = 1 when CN5 is used for the parallel mode
		Bit3: Parallel implemented by CAN2	Bit = 1 when CAN2 is used for parallel control
		Bit4: Group control with CTRL80(M)	Bit = 1 when the CTRL80(M) is involved in parallel mode or group mode.

Note

By default, the CAN1 communication port is used for parallel control so that the CTRL80(M2) can be seamlessly compatible with the CTRL80(M). Therefore, you need not select the parallel control port. When the CAN2 communication port is used for parallel control, you need not set the CTB address switch.

5.1.2 Setting of Physical Floors

Physical floors, relative to the CTRL80 control system, are defined by the installation position of the leveling flag. The floor (such as the ground floor) at which the lowest leveling flag is installed corresponds to physical floor 1. The top physical floor is the accumulative number of the leveling flags. In two-elevator parallel mode, the physical floor numbers of the same floor for two elevators are consistent.

If the floor structures for two elevators are different, physical floors should start with the lowest leveling flag installation position. The physical floors at the overlapped area of the

two elevators are the same. Even if one elevator does not stop a floor in the overlapped area, a leveling flag should be installed there. You can make the elevator not stop at the floor by setting service floors.

When two elevators are in parallel mode, the addresses of the HCBs should be set according to physical floors. Parallel running can be implemented only when the HCB addresses of one floor are set the same for the two elevators.

Note

In parallel mode, the top floor (F6-00) and bottom floor (F6-01) of the elevators should be set based on corresponding physical floors.

Assume that there are two elevators in parallel. Elevator 1 stops at floor B1, floor 1, floor 2, and floor 3, while elevator 2 stops at floor 1, floor 3, and floor 4. Now, you need to set related parameters and HCB addresses according to the following table.

Table 5-2 Parameter setting and HCB addresses of two elevators

		Elevator 1		Elevator 2	
Number of elevators in group mode (FD-00)		2		2	
Elevator SN (FD-01)		1		2	
Actual floor	Physical floor	HCB address	HCB display	HCB address	HCB display
B1	1	1	FE-01 = 1101		
1	2	2	FE-02 = 1901	2	FE-02 = 1901
2	3	3	FE-03 = 1902	Bypass floor but leveling flag required	FE-03 = 1902
3	4	4	FE-04 = 1903	4	FE-04 = 1903
4	5			5	FE-05 = 1904
Bottom floor (F6-01)		1		2	
Top floor (F6-00)		4		5	
Service floor (F6-05)		65535		65531 (the elevator does not stop at physical floor 3)	

5.1.3 Group Mode

The group control board is required if three or more elevators work in group mode. For details, contact IFE.

5.2 Emergency Evacuation at Power Failure

Passengers may be trapped in the car if power failure suddenly happens during the use of the elevator. The evacuation function at power failure is designed to solve the problem.

The evacuation function is implemented in the following two modes:

- Uninterrupted power supply (UPS)
- Shorting PMSM stator

Here we first introduce the implementation through the CTRL80's shorting PMSM stator scheme, which can save energy and improve safety.

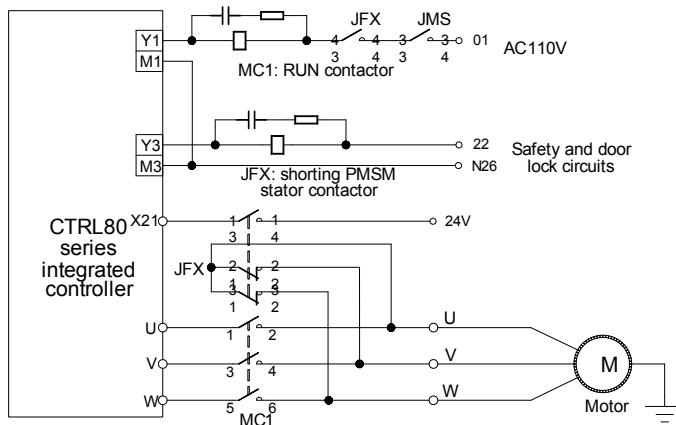
5.2.1 Shorting PMSM Stator

Shorting PMSM stator means shorting phases UVW of the PMSM, which produces resistance to restrict movement of the elevator car. In field application, an auxiliary NC contact is usually added to the NO contact of the output contactor to short UVW phases of the PMSM to achieve the effect. It is feasible in theory but may cause over-current actually. Due to the poor quality of the contactor and the wiring of adding the auxiliary contact, the residual current of the controller is still high when the outputs UVW are shorted at abnormal stop. This results in an over-current fault and may damage the controller or motor.

In our scheme, an independent contactor for shorting PMSM stator and a relay NC contact are used. On the coil circuit of the RUN contactor, an NO contact of the shorting PMSM stator contactor is connected in serial to ensure that output short-circuit does not occur when the parameter setting is incorrect.

The following figure shows wiring of the independent shorting PMSM stator contactor.

Figure 5-1 Wiring of the independent shorting PMSM stator contactor

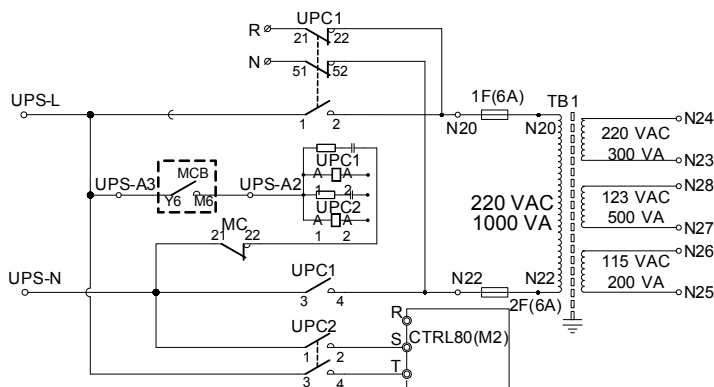


In the wiring shown in the preceding figure, you need to allocate the output terminal of the shorting PMSM stator contactor (Y3, F5-28) with function 12 (Shorting PMSM stator contactor). Set Bit8 of FE-33 (Bit8 = 1, indicating NC output of shorting motor stator contactor; Bit8 = 0, indicating NO output of shorting motor stator contactor) based on the NO/NC state of the connected contact. In addition, connect an input terminal to the X side for monitoring the working status of the shorting motor stator contactor, and allocate this input terminal with function 30. As shown in the preceding figure, X21 is used to monitor the shorting motor stator contactor, and therefore, F5-21 is set to 30.

5.2.2 Emergency Evacuation 220 V UPS

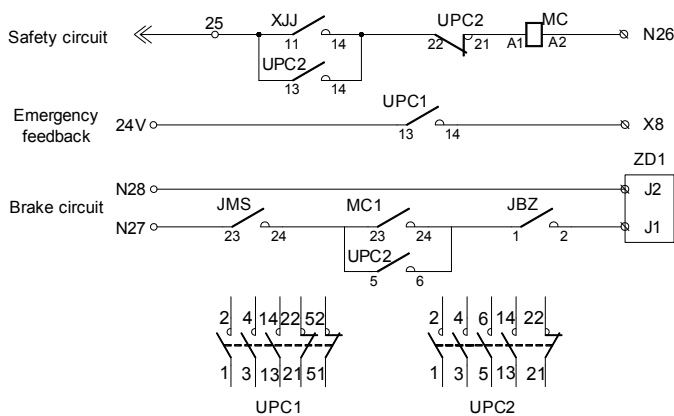
In this scheme, the 220 V UPS provides power supply to the main unit and the drive control circuit. The following figure shows the emergency evacuation 220 V UPS circuit.

Figure 5-2 Emergency evacuation 220 V UPS circuit



The following figure shows various contacts of the contactors.

Figure 5-3 Various contacts of the contactors



The UPS power is recommended in the following table.

Table 5-3 Recommended UPS power for each power rating

UPS Power	Controller Power
1 kVA (700–800 W)	$P \leq 5.5 \text{ kW}$
2 kVA (1400–1600 W)	$5.5 \text{ kW} < P \leq 11 \text{ kW}$
3 kVA (2100–2400 W)	$P \geq 15 \text{ kW}$

The following table lists the setting of the related parameters.

Table 5-4 Parameter setting under the 220 V UPS scheme

Function Code	Setting Value	Description
F5-08 (X8)	27	UPS valid signal
F5-31 (Y6)	13	Emergency evacuation automatic switchover
F6-48	0.010–0.630 m/s	Emergency evacuation switching speed
F6-49	0 to F6-01	Evacuation parking floor
F8-09	0.05 m/s	Emergency evacuation operation speed at power failure
F8-10	0: Motor not running 1: UPS 2: 48 V battery power supply	Emergency evacuation operation mode at power failure

More details on the emergency evacuation setting are provided in F6-45, as listed in the following table.

Table 5-5 Parameter description of F6-45

Bit	Function Description	Binary Setting						Remarks	
Bit0	Direction determine mode	0	Automatically calculating the direction	0	Half-load direction determine	1	Nearby stop	If the torque direction is automatically calculated, the no-load cell function must be enabled, that is, F8-01 is set to 2.	
Bit1		0		1		0			
Bit2	Stop position	1	Stop at the base floor						
		0	Stop nearby						
Bit3	Single leveling door open	1	Stop and open door at a single leveling signal						
		0	Normal leveling stop						
Bit4	Startup compensation	1	Startup torque compensation valid in emergency evacuation running						When it is set that the torque direction is automatically calculated, enable automatic startup torque compensation.

Bit	Function Description	Binary Setting		Remarks
Bit8	Emergency evacuation running time protection	1	If the elevator does not arrive at the required floor after 50s emergency evacuation running time, E31 is reported.	This function is disabled when the function of bit12 (Shorting stator braking mode switched over to controller drive) is enabled.
Bit10	Emergency buzzer output	1	The buzzer output is active during UPS emergency evacuation running.	
Bit12	Shorting stator braking mode switched over to controller drive	1	Enable the function of switching over the shorting stator braking mode to controller drive.	
Bit13	Mode of shorting stator braking mode switched over to controller drive	1	Speed setting	If the speed is still lower than the value set in F6-48 after the elevator is in shorting stator braking mode for 10s, the controller starts to drive the elevator.
		0	Time setting	If the time of the shorting stator braking mode exceeds 50s, the controller starts to drive the elevator.
Bit14	Emergency evacuation exit mode	1	Exit at door close limit	
		0	Exit at door open limit	
Bit15	Function selection of shorting stator braking mode	1	Enable this function.	When this function is enabled, the setting of related function codes becomes effective.

Note

The definition of shorting stator braking mode (for PMSM) is as follows:

Upon power failure, UPS is used, the motor stator is shorted, and the brake is automatically released, making the car move slowly under the effect of the weighing difference between the car and the counterweight.

5.3 Opposite Door Control

Set related parameters according to the following table.

Table 5-6 Parameter setting for opposite door control

Function Code	Parameter Description	Setting Range	Default
Group F5	69/101: Back door forbidden	Function codes of inputs available on the MCB	None
F6-40	Program control selection 1	Bit0: Disability function	0
		Bit2: JP16 input used as the back door selection	0
		Bit3: JP16 input used as the back door open signal	0
		Bit4: Only one door of opposite doors opened under manual control	0
		Bit12: Car call assisted command in single door used as disability function	0
		Bit13: Folding command used as disability function when Bit13 is 1 and back door function when Bit13 is 0	0
		Bit14: Car call command folding	0
		Bit15: JP20 used for switchover to back door	0
FB-00	Number of door machine(s)	1-2	1
Group FD	3: Door 2 selection 4: Door 2 restriction 7: Single/Double door selection	Function codes for inputs available on the HCB-B parallel/serial conversion board	None
FE-33	Elevator function selection	Bit15: Opposite door independent control	0

Table 5-7 Selection of the opposite door mode

Opposite Door Mode	CCB Connection Method	Related Parameter Setting
Old mode	The front door CCB is connected to CN7 on the CTB. CN2 on the back door CCB is cascaded to CN1 on the front door the CCB.	FE-33 Bit15 = 1, FB-00 = 2 F6-40 Bit13 = 0, Bit14 = 1
	CN2s on the front door CCB and back door CCB are respectively connected to CN7 and CN8 on the CTB.	FE-33 Bit15 = 1, FB-00 = 2 F6-40 Bit13 = 1, Bit14 = 1

Opposite Door Mode	CCB Connection Method	Related Parameter Setting
New mode	The front door CCB is connected to CN7 on the CTB. CN2 on the back door CCB is cascaded to CN1 on the front door the CCB.	FB-00 = 2, F8-16 = N (N > present top floor) F6-40 Bit13 = 0, Bit14 = 1
	CN2s on the front door CCB and back door CCB are respectively connected to CN7 and CN8 on the CTB.	FB-00 = 2, F8-16 = N (N > present top floor) F6-40 Bit14 = 0 or FB-00 = 2 F8-16 = N (N > present top floor) F6-40 Bit13 = 1, Bit14 = 1

Note

The differences between the old mode and the new mode are as follows:

1. In the old mode, Bit15 of FE-33 needs to be set to 1, and such setting is not required in the new mode.
 2. The hall call address of the back door is set differently in two modes. In the old mode, the hall call address of the back door is set to floor number+16 (used in CTRL80(M)). In the new mode, the hall call address of the back door is set by means in F8-16 (Start address of hall call assisted command). If the present floor is 8, F8-16 can be set to a number larger than 8. If F8-16 is set to N, the hall call address of the back door is present floor number+ N.
-
-

Table 5-8 Use of the opposite door control mode

Opposite Door Control Mode	Function Description	HCB-B Assistance Setting
DFC-04 = 0	Simultaneous control on the opposite door: The front door and back door act simultaneously. Both door open after arrival regardless of front door call or back door call.	If "Door 2 selection" is valid (FD-11 = 3), this does not affect the normal functions.
		If "Door 2 restriction" is valid (FD-12 = 4), only the front door opens.
		If "Door 2 selection" is valid (FD-11 = 3) and "Door 2 restriction" is valid (FD-12 = 4), only the front door opens.
FC-04 = 1	Hall call independent, car call simultaneous: The front door opens upon arriving at the floor with front door hall call. The back door opens upon arriving at the floor with back door hall call. Both doors open if the two doors have hall call respectively or there is car call.	If "Door 2 selection" is valid (FD-11 = 3), this does not affect the normal functions.
		"Door 2 restriction" is valid (FD-12 = 4), and only the front door opens.
		If "Door 2 selection" is valid (FD-11 = 3) and "Door 2 restriction" is valid (FD-12 = 4), only the front door opens.

Opposite Door Control Mode	Function Description	HCB-B Assistance Setting
FC-04 = 2	Hall call independent, opposite-door manual control: The hall call is independent. The front door opens upon arriving at the floor with front door hall call. The back door opens upon arriving at the floor with back door hall call. Both doors open if the two doors have hall call respectively. The front door and back door are controlled by JP16 (F6-40 Bit2 = 1) or JP20 (F6-40 Bit15 = 1) for arriving at the floor with car call.	If "Single/Double door selection" is invalid (FD-11 = 7), the single door open mode is used, indicating that only one door opens upon arriving at the floor required by the car call. If "Only one door of opposite doors open under manual control" is valid (F6-40 Bit4 = 1), only one door opens in any condition, and the other door keeps closed. If "Only one door of opposite doors open under manual control" is invalid (F6-40 Bit4 = 0), the other door can be opened after one door opens and back door switchover is performed upon door open limit.
		If "Single/Double door selection" is valid (FD-11 = 7), the double door control mode is used, indicating that both doors open upon arriving at the floor required by the car call. It is the same as opposite door control mode 1.
		If "Door 2 restriction" is valid (FD-12 = 4), only the front door opens.
		If "Door 2 selection" is valid (FD-11 = 3), only the back door opens.
		If "Door 2 selection" is valid (FD-11 = 3) and "Door 2 restriction" is valid (FD-12 = 4), only the front door opens.
FC-04 = 3	Hall call independent, car call independent: The front door open at front door call, and the back door open at back door call. Both doors open if the two doors have call respectively.	If "Door 2 selection" is valid (FD-11 = 3), only the back door opens.
		If "Door 2 restriction" is valid (FD-12 = 4), only the front door opens.
		If "Door 2 selection" is valid (FD-11 = 3) and "Door 2 restriction" is valid (FD-12 = 4), only the front door opens.

5.4 Disability Function

The function codes related to the disability function are listed in the following table.

Table 5-9 Function codes related to the disability function

Function Code	Parameter Description	Setting Range	Default
F6-40	Program control selection 1	Bit0: Disability function	0
		Bit12: Car call assisted command in single door used as disability function	
		Bit13: Folding command used as disability function when Bit13 is 1 and back door function when Bit13 is 0	
		Bit14: Car call command folding	
FB-15	Special door open holding time	10–1000s	30s

F6-40 Bit0 = 1: The disability function is enabled.

The car call setting is described as follows:

- F6-40 Bit14 = 1: Command folding is enabled.
- F6-40 Bit13 = 1: The folding commands are used for disability calls. The first CCB (floors 1 to 16) connected to CN7 or CN8 is used for common calls, and the cascaded second CCB (floors 17 to 32) are used for disability calls.
- F6-40 Bit13 = 0: The folding commands are used for back door calls.
- F6-40 Bit14 = 0: Command folding is disabled.
- F6-40 Bit12 = 1: The CCB connected to CN7 is used for common calls, and the CCB connected to CN8 is used for disability calls. A total of 40 floors are supported.
- F6-40 Bit12 = 0: The CCB connected to CN7 is used for front door calls, and the CCB connected to CN8 is used for back door calls. A total of 40 floors are supported.

The hall call setting is described as follows:

- F6-40 Bit14 = 1: Command folding is enabled.
- F6-40 Bit13 = 1: The HOP with the address larger than 16 is used for disability calls (only one door).
- F6-40 Bit13 = 0: The HCB-B is used as the call signal input device. The opposite door is supported. The HCB-B with the address smaller than 16 is used for front door calls, and the HCB-B with the address larger than 16 is used for back door calls. JP5 and JP6 on the HCB-B are used for disability input.
- F6-40 Bit14 = 0: Command folding is disabled. The RS232 serial port on the MCB is used for disability input, and the address setting is consistent with the actual floors. A total of 40 floors are supported.

Note

CN8 is used only for disability or back door calls. For single door control or front door car calls, do not use CN8.



6

Troubleshooting

Chapter 6 Troubleshooting

6.1 Description of Fault Levels

The CTRL80(M2) controller has more than 60 pieces of alarm information and protective functions. It monitors various input signals, running conditions and feedback signals. If a fault occurs, the relevant protective function acts and the system displays the fault code.

The controller is a complicated electronic control system and the displayed fault information is graded into five levels according to the severity. The faults of different levels are handled according to the following table.

Table 6-1 Fault levels

Category	Action	Remarks
Level 1	Display fault code. Output fault relay action command.	1A: The elevator running is not affected on any condition.
Level 2	Display fault code. Output fault relay action command. Continue normal running of the elevator.	2A: The parallel control function is disabled.
		2B: The pre-open/re-leveling function is disabled.
Level 3	Display fault code. Output fault relay action command. Block output and apply the brake immediately after stop.	3A: In low-speed running, the elevator stops at special deceleration rate, and cannot restart.
		3B: In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.
Level 4	Display fault code. Output fault relay action command. In distance control, the elevator decelerates to stop and cannot run again.	4A: In low-speed running, the elevator stops under special deceleration rate, and cannot restart.
		4B: In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.
		4C: In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.
Level 5	Display fault code. Output fault relay action command. The elevator stops immediately.	5A: In low-speed running, the elevator stops immediately and cannot restart.
		5B: In low-speed running, the elevator does not stop. In normal-speed running, the elevator stops, and then can start running at low speed after a delay of 3s.

Note

- A, B, and C are fault sub-category.
 - Low-speed running involves inspection, emergency evacuation, shaft auto-tuning, re-leveling, motor auto-tuning, base floor detection, and running in operation panel control.
 - Normal-speed running involves automatic running, returning to base floor in fire emergency state, firefighter operation, attendant operation, elevator locked, and elevator parking
-
-

6.2 Fault Information and Troubleshooting

If an alarm is reported, the system performs corresponding processing based on the fault level. You can handle the fault according to the possible causes described in the following table.

Table 6-2 Fault codes and troubleshooting

Fault Code	Name	Possible Causes	Solution	Level
E02	Over-current during acceleration	1. The main circuit output is grounded or short circuited. 2. Motor auto-tuning is performed improperly. 3. The load is too heavy. 4. The encoder signal is incorrect.	1. Check whether the RUN contactor at the controller output side is normal. 2. Check: • Whether the power cable jacket is damaged • Whether the power cable is possibly short circuited to ground • Whether the cable is connected reliably 3. Check the insulation of motor power terminals, and check whether the motor winding is short-circuited or grounded. 4. Check whether the shorting PMSM stator contactor causes controller output short circuit. 5. Check whether motor parameters comply with the nameplate. 6. Perform motor auto-tuning again. 7. Check whether the brake keeps released before the fault occurs and whether the brake is stuck mechanically. 8. Check whether the balance coefficient is correct. (To be continued)	5A
E03	Over-current during deceleration	1. The main circuit output is grounded or short circuited. 2. Motor auto-tuning is performed improperly. 3. The load is too heavy. 4. The deceleration rate is too short. 5. The encoder signal is incorrect.	3. Check the insulation of motor power terminals, and check whether the motor winding is short-circuited or grounded. 4. Check whether the shorting PMSM stator contactor causes controller output short circuit. 5. Check whether motor parameters comply with the nameplate. 6. Perform motor auto-tuning again. 7. Check whether the brake keeps released before the fault occurs and whether the brake is stuck mechanically. 8. Check whether the balance coefficient is correct. (To be continued)	5A

Fault Code	Name	Possible Causes	Solution	Level
E04	Over-current at constant speed	1. The main circuit output is grounded or short circuited. 2. Motor auto-tuning is performed properly. 3. The load is too heavy. 4. The encoder is seriously interfered with.	9. Check whether the encoder wirings are correct. For asynchronous motor, perform SFVC and compare the current to judge whether the encoder works properly. 10. Check: • Whether encoder pulses per revolution is set correctly • Whether the encoder signal is interfered with • Whether the encoder cable runs through the duct independently • Whether the cable is too long • Whether the shielding layer is grounded at one end 11. Check: • Whether the encoder is installed reliably • Whether the rotating shaft is connected to the motor shaft reliably • Whether the encoder is stable during normal-speed running 12. Check whether UPS feedback is valid in the non-UPS running state (E02). 13. Check whether the acceleration/deceleration rate is too high. (End)	5A

Fault Code	Name	Possible Causes	Solution	Level
E05	Over-voltage during acceleration	1. The input voltage is too high. 2. The regeneration power of the motor is too high. 3. The braking resistance is too large, or the braking unit fails. 4. The acceleration rate is too short.	1. Adjust the input voltage. Observe whether the bus voltage is normal and whether it rises too quickly during running. 2. Check for the balance coefficient. 3. Select a proper braking resistor and check whether the resistance is too large based on the recommended braking resistance table in chapter 1.	5A
E06	Over-voltage during deceleration	1. The input voltage is too high. 2. The braking resistance is too large, or the braking unit fails. 3. The deceleration rate is too short.	4. Check: - Whether the cable connecting the braking resistor is damaged - Whether the cooper wire touches the ground - Whether the connection is reliable	5A
E07	Over-voltage at constant speed	1. The input voltage is too high. 2. The braking resistance is too large, or the braking unit fails.		5A
E08	Power-on cache relay not energized	The drive control board fails.	Replace the controller.	5A
E09	Under-voltage	1. Instantaneous power failure occurs on the input power supply. 2. The input voltage is too low. 3. The drive control board fails.	1. Eliminate external power supply faults and check whether the power fails during running. 2. Check whether the wiring of all power input cables is reliable. 3. Contact the agent or IFE.	5A

Fault Code	Name	Possible Causes	Solution	Level
E10	Controller overload	1. The brake circuit is abnormal. 2. The load is too heavy. 3. The encoder feedback signal is abnormal. 4. The motor parameters are incorrect. 5. A fault occurs on the motor power cables.	1. Check the brake circuit and power input. 2. Reduce the load. 3. Check whether the encoder feedback signal and setting are correct, and whether the initial angle of the encoder for the PMSM is correct. 4. Check the motor parameter setting and perform motor auto-tuning. 5. Check the power cables of the motor (refer to the solution of E02). 6. Contact IFE.	4A
E11	Motor overload	1. FC-02 is set improperly. 2. The brake circuit is abnormal. 3. The load is too heavy.	1. Adjust the parameter (FC-02 can be set to the default value). 2. Refer to E10.	3A
E12	Power supply phase loss	1. The power input is not symmetric. 2. The drive control board fails.	1. Check whether the three phases of power supply are balanced and whether the power voltage is normal. If not, adjust the power input. 2. Contact IFE.	4A
E13	Power output phase loss	1. The output wiring of the main circuit is loose. 2. The motor is damaged.	1. Check the wiring. 2. Check whether the contactor at the output side is normal. 3. Eliminate the motor fault.	4A
E14	Module overheat	1. The ambient temperature is too high. 2. The fan is damaged. 3. The air filter is blocked.	1. Lower the ambient temperature. 2. Clear the air channel. 3. Replace the damaged fan. 4. Check whether the installation clearance of the controller satisfies the requirement.	5A
E16	Current control fault	1. The output air switch becomes off. 2. The values of the current loop parameters are too small. 3. The zero-point position is incorrect. 4. The load is too heavy.	1. Check the circuit of the encoder. 2. Check the output circuit and grounding cable. 3. Adjust the values of the current loop parameters. 4. Perform motor auto-tuning again. 5. Adjust the balance coefficient.	5A

Fault Code	Name	Possible Causes	Solution	Level
E17	Reference signal incorrect	1. The Z signal of the encoder is interfered with. 2. A hardware fault occurs on the circuit of the encoder.	1. Check whether the encoder runs properly. 2. Check whether the encoder wiring is correct and reliable. 3. Check whether the PG card wiring is correct. 4. Check whether the grounding is reliable.	5A
E18	Current detection fault	The drive control board fails.	Replace the controller.	5A
E19	Motor auto-tuning fault	1. The motor cannot rotate properly. 2. The motor auto-tuning times out. 3. The encoder for the PMSM fails.	1. Enter the motor parameters correctly. 2. Check the motor wiring and whether phase loss occurs on the contactor at the output side. 3. Check the feedback encoder wiring and ensure that encoder pulses per revolution of the encoder is set properly. 4. Check whether the brake keeps released during no-load auto-tuning. 5. Check whether the inspection button is released before the PMSM with-load auto-tuning is finished.	5A
E20	Encoder fault	1. The encoder model does not match the motor. 2. The encoder wiring is incorrect. 3. The current keeps large during low-speed running.	1. Check whether F1-00 is set correctly. 2. Check the encoder wiring. 3. Check whether the encoder power supplied by the PG card is normal if E20 is reported in motor auto-tuning and stop states for UVW encoder. 4. Check whether the encoder is stuck mechanically during running. 5. Check whether the brake keeps released during running.	5A
E22	Leveling signal abnormal	In automatic running state, if the floor pulse error is too large, the elevator returns to end floor for check.	1. Check whether the leveling and door zone sensors work properly. 2. Check the installation verticality and depth of the leveling flags. 3. Check the input points of the MCB.	1A

Fault Code	Name	Possible Causes	Solution	Level
E24	System clock fault	The real-time clock (RTC) storage is abnormal.	Replace the controller.	3B
E25	EEPROM failure	EEPROM of the I2C (inter-integrated circuit) times out.	1. Power off the controller, and power it again. 2. Replace the controller.	4A
E26	Earthquake signal	1. An earthquake happens. 2. The earthquake sensor mal-acts. 3. The earthquake signal input parameter is set incorrectly.	1. Stay calm and find a place that can form triangular space to hide. 2. Check whether the earthquake sensor is reliable and whether the state (NO, NC) of the feedback contact is consistent with the parameter setting of the MCB.	3B
E29	Shorting PMSM stator feedback abnormal	1. The input parameter for shorting PMSM stator feedback is set incorrectly. 2. The wiring between the shorting PMSM stator contactor and the feedback circuit is incorrect.	1. Check whether the state (NO, NC) of the feedback contact on the contactor is consistent with the parameter setting of the MCB. 2. Check whether the state of the MCB output indicator is consistent with the contactor action. 3. Check whether corresponding feedback contact acts after the contactor acts, and whether the corresponding feedback input point of the MCB acts correctly. 4. Check whether the shorting PMSM stator contactor is consistent with the MCB output feature. 5. Check the coil circuit of the shorting PMSM stator contactor.	3B
E30	Elevator position abnormal	1. The controller does not receive leveling signal within the time set in F9-02. 2. The up and down limit switches are met during the re-leveling process.	1. Check whether the leveling sensor acts correctly. 2. Check whether the leveling signal cables are connected reliably and whether the signal copper wires may touch the ground or be short circuited with other signal cables. 3. Check whether the encoder is used properly. 4. Check whether the limit switches act properly.	4A

Fault Code	Name	Possible Causes	Solution	Level
E33	Elevator speed abnormal	1. The running speed exceeds the limit in distance control mode. 2. The speed exceeds the limit during self-sliding. 3. The speed exceeds the limit or re-leveling is not performed within the required time during emergency running.	1. Check whether the encoder is used properly. 2. Check the setting of motor nameplate parameters. Perform motor auto-tuning again. 3. Check whether the inspection signal is valid at normal speed running. 4. Check the inspection switch and signal cables. 5. Check the leveling signal.	5A
E34	Logic fault	Logic of the MCB is abnormal	Contact IFE to replace the MCB.	4C
E35	Shaft auto-tuning data abnormal	1. The elevator is not at the bottom floor when shaft auto-tuning is started. 2. No leveling signal is received within 45s while continuous running. 3. The distance between two floors is too small. 4. The maximum number of landing floors is inconsistent with the setting value. (To be continued)	1. Upon power-on, E35 is reported when the flag height is detected to be 0. In this case, check whether the data obtained after shaft auto-tuning is changed, resulting too large check error. 2. If E35 is reported when the RUN contactor is not energized, check: • Whether the next slow-down switch is valid • Whether F4-01 (Current floor) is set to 1 • Whether the inspection switch is in inspection state and supports inspection running • Whether F0-00 is set to 1 (To be continued)	4C

Fault Code	Name	Possible Causes	Solution	Level
E35	Shaft auto-tuning data abnormal	5. The floor pulses change inversely. 6. The system is not in the inspection state when shaft auto-tuning is performed. 7. Shaft auto-tuning is not performed upon power-on or the check data error is too large. 8. The down slow-down limit is invalid or the control mode is SFVC. (End)	3. If E35 is reported at the first leveling position, check: - Whether the value of F4-03 increases in UP direction. If not, change the value of F2-10 - Whether the NO/NC setting of the leveling sensor is set correctly - Whether the leveling flags are inserted properly if the leveling sensor signal blinks 4. If E35 is reported during running, check: - Whether the running times out: No leveling signal is received when the running time exceeds F9-02 - Whether the super short floor function is enabled when the floor distance is less than 500 mm - Whether the setting of F6-00 (Top floor of the elevator) is smaller than the actual condition 5. If E35 is reported when the elevator arrives at the top floor, check: - Whether the obtained top floor of the elevator and bottom floor of the elevator are consistent with the setting of F6-00 and F6-01 when the up slow-down signal is valid and the elevator reaches the door zone - Whether the obtained floor distance is less than 500 mm (End)	

Fault Code	Name	Possible Causes	Solution	Level
E36	RUN contactor feedback abnormal	1. The running output is inconsistent with the feedback signal. 2. The feedback contacts of the RUN contactor are in different states. 3. The running current of the elevator is less than 0.1 A after the output of both the RUN contactor and the brake contactor.	1. Check whether the feedback contact of the contactor acts properly. 2. Check whether the feedback contacts (NO, NC) of the contactor are consistent with the MCB parameter setting. 3. Check whether the output cables UVW of the controller are connected properly. 4. Check whether the power supply of the controller's control circuit is normal. 5. Check whether the states and parameter settings of all contacts are correct.	5A
E37	Brake contactor feedback abnormal	1. The output of the brake contactor is inconsistent with the feedback. 2. The feedback contacts of the RUN contactor are in different states.	1. Check whether the brake coil and feedback contact are correct. 2. Check the signal feature (NO, NC) of the feedback contact. 3. Check whether the power supply of the brake coil's control circuit is normal.	5A
E38	Encoder signal abnormal	1. There is no input of the encoder pulses when the elevator runs automatically. 2. The direction of the input encoder signal is incorrect when the elevator runs automatically. 3. F0-00 is set to 0 (SFVC) in distance control.	1. Check whether the encoder is used correctly. 2. Exchange phases A and B of the encoder. 3. Check the setting of F0-00, and change it to "CLVC". 4. Check whether the system and signal cables are grounded reliably. 5. Check whether cabling between the encoder and the PG card is correct.	5A
E39	Motor overheat	Motor overheat input is valid.	1. Check whether the overheat feedback is reliable and whether the parameter setting is correct. 2. Check whether the motor is used properly and whether it is damaged. 3. Improve cooling conditions of the motor.	3A
E40	Elevator running timeout	The setting of the elevator running time is exceeded.	The elevator is used for a long time and needs maintenance.	4B

Fault Code	Name	Possible Causes	Solution	Level
E41	Safety circuit breaking	1. The safety feedback of the low-voltage detection point is invalid. 2. The safety feedback of the high-voltage detection point is invalid.	1. Check the status of the safety circuit switches. 2. Check whether the external power supply is normal. 3. Check whether the safety circuit contactor acts properly. 4. Confirm the signal feature (NO, NC) of the feedback contact of the safety circuit contactor.	5A
E42	Door lock breaking off during running	The door lock circuit feedback breaks off during the elevator running.	1. Check whether the landing door lock and the car door lock are in good contact. 2. Check whether the door lock contactor acts properly. 3. Check the signal feature (NO, NC) of the feedback contact on the door lock contactor. 4. Check whether the external power supply is normal.	5A
E43	Up limit signal breaking off during running	The up limit signal breaks off when the elevator is running up.	1. Check the signal feature (NO, NC) of the up limit switch and down limit switch. 2. Check whether the up limit and down limit switches act properly.	4C
E44	Down limit signal breaking off during running	The down limit signal breaks off when the elevator is running down.	3. Check whether the installation positions of limit switches are too close to the leveling flag. If so, the limit switches will be touched at normal leveling.	4C
E45	Slow-down switch position abnormal	1. The installation positions of the slow-down switches do not satisfy the slow-down requirements. 2. The recorded slow-down switch position is greatly different from the actual position.	1. Check whether the up slow-down 1 and the down slow-down 1 are in good contact. 2. Check the signal feature (NO, NC) of the up slow-down 1 and the down slow-down 1. 3. Calculate the correct installation positions of the slow-down switches based on the rated elevator speed.	4B

Fault Code	Name	Possible Causes	Solution	Level
E46	Re-leveling abnormal	1. The re-leveling running speed exceeds 0.1 m/s. 2. The elevator is out of the door zone when re-leveling. 3. The feedback of the shorting door lock circuit contactor is abnormal.	1. Check the original and secondary wiring of the shorting door lock circuit relay. 2. Check whether the shorting door lock circuit contactor feedback function is enabled and whether the feedback signal is normal. 3. Check whether the encoder is used properly.	2B
E47	Shorting door lock circuit contactor feedback abnormal	1. The feedback of the shorting door lock circuit contactor is abnormal. 2. The elevator runs at over speed or the running times out when the shorting door lock circuit contactor has output.	1. Check the signal feature (NO, NC) of the feedback contact on the shorting door lock circuit contactor. 2. Check whether the shorting door lock circuit contactor acts properly.	2B
E48	Door open fault	The consecutive times that the door does not open/close to the limit reaches the setting in FB-09.	1. Check whether the door machine system works properly. 2. Check whether the CTB is normal.	5A
E49	Door close fault			5A
E51	CAN communication abnormal	Feedback from the CTB is abnormal.	1. Check the communication cable connection. 2. Check the power supply of the CTB. 3. Check whether the 24 V power supply of the controller is normal.	1A
E52	HCB communication abnormal	Feedback from the HCB is abnormal.	1. Check the communication cable connection. 2. Check whether the 24 V power supply of the controller is normal. 3. Check whether the HCB addresses are repeated.	1A
E53	Shorting door lock circuit fault	1. The back door lock is not released after a certain period of door open signal output. 2. The locks are in different states.	1. Check whether the door lock circuit is normal. 2. Check whether the feedback contact of the door lock contactor acts properly. 3. Check whether the system receives the door open limit signal when the door lock signal is valid.	5A

Fault Code	Name	Possible Causes	Solution	Level
E54	Over-current at startup of inspection running	The current exceeds 110% of the rated current at the startup of inspection running.	1. Reduce the load. 2. Adjust the balance coefficient.	5A
E55	Stop at another landing floor	During automatic running of the elevator, the door open limit signal is not received after the door open time exceeds the protection time	1. Check whether the door open limit signal is reliable. 2. Check whether the door open limit parameters are set correctly.	1A
E56	Door close limit fault	1. The time when both door open limit and door close limit are valid exceeds 2s. 2. Door open limit is achieved 1s after stop.	1. Check whether the door open limit switch is connected and acts reliably. 2. Check whether the NO/NC setting of the door open and close switches is correct. 3. Check whether mal-action exists.	1A
E57	Serial peripheral interface (SPI) communication abnormal	Communication between the control board and the drive board is abnormal.	1. Check the wiring between the control board and the drive board. 2. Check whether a hardware fault occurs, and replace the controller if so.	5A
E58	Shaft position switches abnormal	1. The up limit and down limit are valid simultaneously. 2. The up slow-down and down slow-down are valid simultaneously.	1. Check whether the states (NO, NC) of the slow-down switches and limit switches are consistent with the parameter setting of the MCB 2. Check whether the slow-down switches and limit switches act reliably. 3. Check whether the slow-down switches and limit switches are not wired.	4B



IFE ELEVATORS CO. , LTD

Jinlong Rd, Jinlong Industrial Zone
Qingxi, Dong Guan, Guang Dong,
China 523660

Tel: +86-769-82078888

Fax: +86-769-87732448

Hotline: 400-6789-443

Email: ife@ifelift.com